

Observational Evidence for a Characteristic Kinematic Scale in the High- ϕ Regime of Spiral Galaxies

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Abstract

The flat rotation curves of disk galaxies have long challenged the standard Λ CDM model, motivating the search for characteristic scales encoded in galactic dynamics. We present observational evidence for a characteristic kinematic scale $c_0 \approx 104.3 \text{ km s}^{-1}$ in the high- ϕ regime of spiral galaxies. The scale is constructed from the operational definition

$$c_0 = \frac{v_{\text{bar}}}{\sqrt{e^{2\phi} - 1}},$$

where $\phi = \ln(v_{\text{obs}}/v_{\text{bar}})$ is a dimensionless field inferred directly from the rotation curves, v_{obs} is the observed rotational velocity, and v_{bar} is the circular velocity expected from the baryonic mass distribution alone. The scale c_0 is defined in the regime where the field ϕ is significant ($\phi \gtrsim 0.2$); in the limit $\phi \rightarrow 0$, the definition becomes singular, corresponding to the case $v_{\text{obs}} = v_{\text{bar}}$.

Using 1222 independent measurements from 115 SPARC galaxies, we obtain a clean sample of 723 points (after a physically motivated quality cut) with a mean $\langle c_0 \rangle = 106.9 \pm 43.2 \text{ km s}^{-1}$. A bootstrap 95% confidence interval, $[103.7, 110.0] \text{ km s}^{-1}$, contains the characteristic value $c_0 = 104.3 \text{ km s}^{-1}$ ($p = 0.112$). Progressive cuts on c_0 reveal a striking convergence: the strictest cuts yield a mean of exactly 104.3 km s^{-1} with $p > 0.98$.

To establish the robustness of this scale, we employ a network of independent tests. Three physical cuts—based on galactocentric radius, the field ϕ , and measurement quality, none of which involve c_0 itself—show that the central value shifts systematically toward the characteristic value when physically well-motivated regions are selected. A galaxy-level cross-validation, repeated across 1000 random partitions, gives a mean difference between training and test sets of -0.11 km s^{-1} , supporting the interpretation that the central value is a general property of the sample. The median of the clean sample, 99.0 km s^{-1} , has a 95% confidence interval $[93.7, 104.8]$ that contains the characteristic value. Synthetic datasets generated from a dark-halo model or randomised velocities do not reproduce the observed concentration of c_0 , supporting the interpretation that the concentration is not a mathematical artifact of the definition.

The scatter in galaxy-averaged c_0 values, ranging from ~ 30 to $\sim 250 \text{ km s}^{-1}$, is organised into a systematic “slope, plateau, and valley” morphology that correlates with the magnitude of the field ϕ . Galaxies with small ϕ lie on the slope, those with intermediate ϕ populate a plateau near $c_0 \approx 104 \text{ km s}^{-1}$, and gas-rich dwarfs with large ϕ occupy a distinct low- c_0 valley. The field ϕ reaches a plateau value $\phi_{\text{plateau}} = 0.429$ in massive galaxies, corresponding to a fixed enhancement factor $e^{2\phi} = 2.357$ that quantitatively reproduces the Radial Acceleration Relation without additional free parameters. These results identify $c_0 \approx 104.3 \text{ km s}^{-1}$ as a new empirical kinematic scale in the high- ϕ regime of spiral galaxies.

Keywords: galaxies: kinematics and dynamics, galaxies: spiral, dark matter, methods: statistical

1. Introduction

The flatness of rotation curves in disk galaxies, first clearly demonstrated by (25) for M31 and subsequently confirmed for hundreds of spirals (26; 29), has stood for half a century as one of the most persistent anomalies in astrophysics. Within the standard Λ CDM cosmological model, these observations are interpreted as evidence for extended halos of non-baryonic dark matter that dominate the gravitational potential at large radii (21; 31; 22). While this paradigm has achieved remarkable

success on cosmological scales, its application to galactic scales has encountered persistent difficulties, including the core-cusp problem, the missing satellites problem, and the continued non-detection of dark matter particles (9; 19; 12; 1).

These tensions have motivated alternative descriptions of galactic dynamics, most notably Milgromian dynamics (MOND; (18)), which modifies Newtonian dynamics below a characteristic acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$. MOND successfully reproduces the flatness of rotation curves and the tight Radial Acceleration Relation (RAR) discovered by (16). The empirical success of a single characteristic scale in galactic dynamics raises a broader question: are there other characteristic scales encoded in the data that have not yet been identified?

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The present work addresses this question from a purely observational standpoint.

We introduce a dimensionless quantity constructed directly from the rotation curves,

$$\phi = \ln\left(\frac{v_{\text{obs}}}{v_{\text{bar}}}\right), \quad (1)$$

where v_{obs} is the observed rotational velocity and v_{bar} is the circular velocity expected from the baryonic mass distribution alone. The quantity ϕ is defined operationally as a scalar function over the galactic disk. From ϕ we construct a characteristic kinematic scale,

$$c_0 = \frac{v_{\text{bar}}}{\sqrt{e^{2\phi} - 1}}, \quad (2)$$

which we refer to throughout as the characteristic kinematic scale of the high- ϕ regime (where ϕ approaches its plateau value ϕ_{plateau}). The particular combination $v_{\text{bar}}/\sqrt{e^{2\phi} - 1}$ is chosen because it has the dimensions of velocity, is defined entirely in terms of the observed quantities v_{obs} and v_{bar} , and is finite and continuous for $\phi > 0$. It is a natural combination of v_{bar} and ϕ that yields a velocity scale defined throughout the high- ϕ regime and isolates the characteristic speed associated with that regime.

The scale c_0 is defined in the regime where the field ϕ is significant ($\phi \gtrsim 0.2$); as $\phi \rightarrow 0$ the denominator vanishes and the definition becomes singular. This limit corresponds to the case $v_{\text{obs}} = v_{\text{bar}}$, which lies outside the domain of the quantity we study.

The scale c_0 is not introduced as a replacement for the acceleration scale a_0 ; it provides a complementary empirical characterization of the same physical regime. Whereas a_0 governs the low-acceleration regime across galaxies, c_0 is constructed directly from the velocity field and is defined only in the high- ϕ regime. The two scales are dimensionally related by $a_0 \sim c_0^2/R_{\text{scale}}$, where R_{scale} is a characteristic galactic length scale. However, they characterize different aspects of the dynamics: a_0 is an acceleration scale derived from the mass distribution, whereas c_0 is a velocity scale constructed from the observed kinematics. The motivation for introducing c_0 is not to replace a_0 , but to explore whether a complementary scale constructed directly from the velocity field reveals features that are not immediately apparent in the acceleration-based description. Whether this perspective adds information beyond what is already encoded in a_0 , and whether the dimensional relation above is theoretically significant, are questions for Paper II.

The central question of this paper is whether c_0 exhibits a stable and reproducible central value across a large, homogeneous sample of spiral galaxies. The SPARC database (13; 14) provides an ideal testbed for this investigation, with 175 galaxies and high-quality rotation curves derived from HI and H α observations, together with accurate 3.6 μm photometry for the baryonic decomposition. After applying objective quality criteria, we retain 1222 independent measurements from 115 galaxies, spanning five orders of magnitude in stellar mass and three orders of magnitude in surface brightness.

The analysis reveals that while individual galaxies display substantial scatter in c_0 (from ~ 30 to $\sim 250 \text{ km s}^{-1}$), this variation is organised into a systematic three-fold morphology that correlates with the magnitude of the field ϕ . Galaxies with small ϕ ($\langle\phi\rangle < 0.25$) lie on a “slope” where c_0 is elevated; galaxies with intermediate ϕ populate a “plateau” where c_0 exhibits a stable central value (found in Section 3 to be near 104.3 km s^{-1}) with a reduced scatter compared to the slope and valley regimes; and gas-rich dwarfs with large ϕ ($\langle\phi\rangle > 0.55$) occupy a distinct low- c_0 “valley”. The scale c_0 is defined at the level of individual data points, using the criterion $\phi > 0.2$, whereas the three-fold morphology is defined at the level of galaxies, using the galaxy-averaged field $\langle\phi\rangle$. The boundaries of these regimes are chosen empirically from the observed distribution of $\langle\phi\rangle$ in the SPARC sample; their precise values are discussed in Section 2. The paper establishes the robustness of the central value through a network of independent tests: progressive cuts on c_0 , physical cuts based on galactocentric radius and the field ϕ (criteria that do not involve c_0 itself), galaxy-level cross-validation (mean difference between training and test sets of -0.11 km s^{-1}), median analysis, and synthetic data tests. Each of these tests is designed to probe a different aspect of the robustness of the scale, and together they provide a comprehensive empirical characterisation.

The structure of this paper is as follows. Section 2 describes the SPARC sample, the selection criteria, and the computation of the physical quantities used throughout the analysis. Section 3 presents the empirical findings, including the stability of c_0 in the high- ϕ regime, the behaviour of the field ϕ , the progressive and physical cuts, the cross-validation and median analyses, and the analysis of the Radial Acceleration Relation. Section 4 discusses the interpretation of c_0 as a new empirical scale, compares the results with the dark matter and MOND paradigms, and addresses potential systematics. Our conclusions are summarised in Section 5.

2. Data and Analysis Methodology

2.1. The SPARC Database and Sample Selection

We use the SPARC (Spitzer Photometry and Accurate Rotation Curves) database (13; 14), the largest publicly available collection of high-quality rotation curves for disk galaxies. The sample comprises 175 galaxies spanning Sa to Irr morphological types, five orders of magnitude in stellar mass, and over three orders of magnitude in surface brightness. For each galaxy, SPARC provides two essential components: (i) high-resolution rotation curves $v_{\text{obs}}(r)$ derived from calibrated 21-cm HI line observations (primarily Westerbork, Very Large Array, and Arecibo telescopes) and, where available, optical H α long-slit spectroscopy, with quoted uncertainties accounting for both measurement errors and asymmetric drift corrections; and (ii) accurate 3.6 μm Spitzer photometry, decomposed into stellar disk, bulge (if present), and gaseous disk (HI + He) contributions to the circular velocity. For each component, $v_{\text{disk}}(r)$, $v_{\text{bulge}}(r)$, and $v_{\text{gas}}(r)$ are computed assuming a thin disk geometry for the stellar disk and gas, and spherical symmetry for

the bulge. The total baryonic circular velocity is obtained by quadrature summation:

$$v_{\text{bar}}(r) = \sqrt{v_{\text{gas}}(r)^2 + v_{\text{disk}}(r)^2 + v_{\text{bulge}}(r)^2}. \quad (3)$$

The complete dataset is publicly available at the SPARC data repository.¹

2.1.1. Sample Selection Criteria

From the initial 175 galaxies, we apply a set of objective quality cuts to ensure the robustness of our analysis. Only galaxies satisfying all of the following criteria are retained:

- Minimum data points: A galaxy must have at least 2 valid measurements after all other cuts are applied, to be included in the final sample. For error estimation purposes, however, a minimum of 10 points per galaxy is required to reliably measure the scatter via the MAD method.
- Physical positivity of ϕ : $v_{\text{obs}} > v_{\text{bar}}$ for each included data point, ensuring $\phi > 0$.
- Numerical stability: $\phi < 2.5$, excluding regions of extreme local dynamics.
- Minimum baryonic velocity: $v_{\text{bar}} > 0.1 \text{ km s}^{-1}$ to avoid division-by-zero instabilities.
- Plausible c_0 range: $30 < c_0 < 300 \text{ km s}^{-1}$. This very wide window is deliberately chosen to avoid biasing the analysis toward any particular value; a subsequent progressive cuts analysis (§3.2) is used to assess the impact of outliers. The fiducial “moderate cut” ($50 < c_0 < 200 \text{ km s}^{-1}$), which isolates the high-quality population in the high- ϕ regime, is motivated empirically in §4.3 and visually justified in Fig. 3, where the excluded points are shown to concentrate in regions of high numerical instability.

Applying all criteria yields a final sample of 115 galaxies with 1222 independent measurements. A complete list of all galaxies with their statistics is provided as machine-readable online material.

2.2. Physical Quantities, Error Estimation, and Systematics

Computation of Physical Quantities. For each data point, we compute the following:

- Dimensionless field ϕ : Inferred directly from the ratio of observed to baryonic velocity,

$$\phi(r) = \ln \left(\frac{v_{\text{obs}}(r)}{v_{\text{bar}}(r)} \right). \quad (4)$$

The quantity ϕ is defined operationally as a scalar function over the galactic disk; its physical interpretation is discussed in Paper II.

- Characteristic kinematic scale c_0 : For each point with $\phi(r) > 0$, we use the operational definition:

$$c_0(r) = \frac{v_{\text{bar}}(r)}{\sqrt{e^{2\phi(r)} - 1}}. \quad (5)$$

This combination is chosen because it has the dimensions of velocity, is constructed entirely from the observed quantities v_{obs} and v_{bar} , and is finite and well-defined throughout the high- ϕ regime ($\phi > 0$). The definition becomes singular in the limit $\phi \rightarrow 0$, which lies outside the domain of the quantity studied here. The central question of this paper—whether c_0 exhibits a stable and reproducible central value in the high- ϕ regime—is tested by examining the statistical distribution of these values across all galaxies and radii.

- Accelerations: The observed centripetal acceleration and the Newtonian acceleration expected from baryonic matter alone are

$$g_{\text{obs}}(r) = \frac{v_{\text{obs}}(r)^2}{r}, \quad g_{\text{N}}(r) = \frac{v_{\text{bar}}(r)^2}{r}, \quad (6)$$

with $1 \text{ kpc} = 3.086 \times 10^{19} \text{ m}$.

Direct Measurement of Random Observational Uncertainties.

A key feature of our analysis is that we do not rely on literature estimates for measurement errors. Instead, we measure them directly from the SPARC data. For each galaxy with at least 10 data points, we estimate the scatter in v_{obs} and v_{bar} about their galaxy-wide medians using the median absolute deviation (MAD), scaled to a Gaussian-equivalent standard deviation (24):

$$\sigma_{v_{\text{obs}}} = 1.4826 \times \text{MAD}(v_{\text{obs}}), \quad \sigma_{v_{\text{bar}}} = 1.4826 \times \text{MAD}(v_{\text{bar}}). \quad (7)$$

The global uncertainty estimates, obtained by taking the median of the galaxy-specific values, are:

$$\sigma_{v_{\text{obs}}} = 8.9 \text{ km s}^{-1}, \quad \sigma_{v_{\text{bar}}} = 9.7 \text{ km s}^{-1}. \quad (8)$$

These are significantly smaller than typical literature values (~ 10 and $\sim 15 \text{ km s}^{-1}$, respectively), indicating that the SPARC data are of higher quality than often assumed. These empirically determined errors serve as the input for the Monte Carlo simulations (§3.4), ensuring that the comparison between expected and observed scatter is as realistic as possible.

Systematic Uncertainties and Sensitivity to M/L . While the above analysis addresses random measurement errors, our results are also subject to systematic uncertainties, most notably in the stellar mass-to-light ratio (M/L) used to convert Spitzer $3.6 \mu\text{m}$ photometry into stellar mass. SPARC adopts a constant $\Upsilon_{3.6} = 0.5 M_{\odot}/L_{\odot}$ for spiral galaxies, with small adjustments for dwarf galaxies. To quantify the impact of this assumption, we perform a sensitivity analysis in Appendix Appendix D, varying M/L by $\pm 20\%$. The resulting change in $\langle c_0 \rangle$ is only 2–3%, well within our statistical confidence interval. We also

¹<https://astroweb.cwru.edu/SPARC>

note that the gas contribution v_{gas} is measured directly from HI observations and is unaffected by M/L uncertainties; in gas-rich galaxies this further reduces the sensitivity of our results to stellar mass modelling.

2.3. Statistical Methods

We employ a comprehensive set of statistical methods to characterise the behaviour of c_0 and to establish the robustness of the kinematic scale it defines. All analyses are implemented in Python 3.10 using NumPy, SciPy, Pandas, and Matplotlib (10; 30; 17; 11). The complete analysis code is publicly available on GitHub.²

Table 1 summarises the statistical tests employed in this work. Each test is designed to probe a different aspect of the robustness of c_0 ; together, they form a network of independent consistency checks.

Bootstrap Confidence Intervals. We compute the 95% confidence interval for the mean of c_0 using bootstrap resampling with 10,000 iterations. This non-parametric method does not assume a Gaussian parent distribution and provides a robust estimate of the uncertainty in the mean.

One-Sample t -Test. We compare the mean of the c_0 distribution against a characteristic value $c_0 = 104.3 \text{ km s}^{-1}$. This value is defined in Appendix Appendix B and corresponds to the central value obtained from the strictest progressive cuts described in §3.2. The t -test serves as an internal consistency check of the empirical scale: the characteristic value is itself derived from the SPARC data, and the test quantifies whether the central value of the clean sample is statistically compatible with it. A non-significant result ($p > 0.05$) indicates that the two values are statistically indistinguishable.

2.3.1. Progressive Cuts Analysis.

To assess the influence of potentially low-quality data points, we examine the mean and 95% confidence interval of c_0 under successively stricter selection windows: $0 < c_0 < 300$, $50 < c_0 < 200$, $70 < c_0 < 150$, $80 < c_0 < 130$, and $85 < c_0 < 125 \text{ km s}^{-1}$. This provides a direct test of whether the result is stable or an artefact of a particular cut. Our fiducial “moderate cut” ($50 < c_0 < 200 \text{ km s}^{-1}$) is chosen *a posteriori* as the broadest window that removes the tails of the distribution while preserving the core; the empirical motivation for this window is discussed in §4.3 and illustrated in Figure 3.

2.3.2. Physical Cuts Based on Independent Criteria.

To verify that the convergence of c_0 is not an artefact of selecting on c_0 itself, we also perform cuts based on physically independent quantities that do not involve c_0 :

1. **Galactocentric radius:** We isolate the outer, high- ϕ regions by retaining only measurements with $R > 2 \text{ kpc}$. This cut is motivated by the observation that points with $R < 2 \text{ kpc}$ suffer from numerical instability due to their small ϕ values (see Fig. 3).

2. **Dimensionless field ϕ :** We retain only points with $\phi > 0.2$, corresponding to the high- ϕ regime where c_0 exhibits its greatest empirical stability.
3. **Measurement quality:** We retain only points whose estimated c_0 error (from error propagation) lies below a prescribed threshold, thereby isolating the most reliable measurements.

The results of these independent cuts are reported in the Results section. They demonstrate that the central value of c_0 shifts systematically toward the characteristic value whenever physically well-motivated regions are selected, confirming that the constancy is a genuine empirical property of the high- ϕ regime.

2.3.3. Cross-Validation on the Galaxy Level.

As a further test of the stability of the central c_0 value, we perform a galaxy-level cross-validation. The 115 galaxies are randomly partitioned into a training set (80%) and a test set (20%). The mean c_0 is computed from the training set, and this value is then compared with the mean of the independent test set. This procedure is repeated 1,000 times with different random partitions. The mean difference between the training and test averages is found to be only -0.11 km s^{-1} , demonstrating that the characteristic scale is a predictable, general property of the sample and not an artefact of a particular subsample.

2.3.4. Median Analysis.

While the arithmetic mean is sensitive to outliers in the tails of the distribution, the median provides a robust measure of the central tendency. We compute the median of the clean sample ($50 < c_0 < 200 \text{ km s}^{-1}$) and construct a 95% bootstrap confidence interval for this statistic. The resulting median of 99.0 km s^{-1} has a confidence interval $[93.7, 104.8] \text{ km s}^{-1}$ that comfortably contains the characteristic value, confirming that the core of the distribution is centred on the characteristic scale.

Monte Carlo Error Simulations. We perform 10,000 realisations of the full dataset, adding Gaussian random errors to v_{obs} and v_{bar} with standard deviations given by Eq. (8). Both uncorrelated and correlated ($\rho = 0.30$, estimated from the data) error models are considered. For each realisation, c_0 is recomputed and its scatter measured, yielding the distribution of scatter expected purely from measurement uncertainties. This is compared with the observed scatter to test for the presence of intrinsic variation. A full description of these simulations, including the correlated error model and detailed results, is provided in Appendix Appendix C.2.

Stratified Analysis. To understand the origin of scatter in c_0 , we bin the data by galactocentric radius and by ϕ , measuring the scatter in each bin. This reveals the systematic behaviour of the uncertainty as a function of physical conditions. The results of this analysis are presented in full in Appendix Appendix C.1.

3. Results

3.1. Final Sample and Stability of c_0

Final Sample. After applying the quality criteria described in §2.1.1, we obtain a final sample of 115 galaxies with a total

²URL to be inserted upon acceptance.

Table 1: Summary of statistical tests applied to the characteristic kinematic scale c_0 .

Test	Purpose	Section
Bootstrap CI	Uncertainty in the mean	§2.3
t -test	Comparison with the characteristic value	§2.3
Progressive cuts	Stability under window variation	§3.2
Physical cuts	Robustness to independent selection	§2.3.2
Cross-validation	Predictability of central value	§2.3
Median analysis	Robustness to outliers	§2.3
Monte Carlo	Scatter vs. measurement errors	§3.4

of 1222 independent measurements. The galaxies span five orders of magnitude in stellar mass, from massive spirals such as NGC 2841 to dwarf irregulars such as F568-1. A representative subset of 15 galaxies is listed in Table 2; the complete statistics for all 115 galaxies are provided as machine-readable online material.

The galaxy-averaged values of c_0 in Table 2 exhibit a substantial range, from ~ 30 to ~ 250 km s $^{-1}$. This scatter is not random but reflects a systematic “slope, plateau, and valley” structure. Galaxies with large values of $\langle c_0 \rangle$ (e.g., NGC 7331, NGC 6946) are those with small $\langle \phi \rangle$ ($\langle \phi \rangle < 0.25$): they lie on the “slope”. Galaxies with intermediate values (e.g., NGC 3769 at 102.4 km s $^{-1}$, NGC 0289 at 108.9 km s $^{-1}$, and NGC 2841 at 124.3 km s $^{-1}$) are on the “plateau”, with several lying very close to the characteristic value $c_0 = 104.3$ km s $^{-1}$. A third class of dwarf galaxies (e.g., UGC 00128, F568-1) have very low c_0 because their baryonic masses are small; they form a distinct “valley”. The central empirical claim of this work is therefore not that c_0 is strictly universal for every galaxy, but that the galaxies in the high- ϕ regime consistently point to the same characteristic scale.

For the full sample of 1222 points, the mean characteristic scale is $\langle c_0 \rangle = 108.7 \pm 72.0$ km s $^{-1}$. However, this distribution contains a substantial tail of extreme values that are likely dominated by measurement uncertainties and numerical instability, as we will demonstrate in §3.4. Applying a moderate quality cut ($50 < c_0 < 200$ km s $^{-1}$), motivated empirically in §3.2 and illustrated in Fig. 3, removes 41% of points and leaves a clean sample of 723 high-quality measurements. For this subset:

$$\langle c_0 \rangle = 106.9 \pm 43.2 \text{ km s}^{-1}. \quad (9)$$

Stability Test. The central empirical test of this work is whether c_0 is consistent with a single stable value in the high- ϕ regime. Figure 1 shows the distribution of bootstrap means for the full and clean samples. For the clean sample, the bootstrap 95% confidence interval for the mean, computed from 10,000 resamplings, is:

$$\text{CI}_{95\%} = [103.71, 110.00] \text{ km s}^{-1}. \quad (10)$$

This interval comfortably contains the characteristic value $c_0 = 104.3$ km s $^{-1}$ (defined in Appendix B). A one-sample t -test against this value yields $p = 0.112$, indicating that the measured mean is statistically indistinguishable from the central value inferred from the strictest progressive cuts.

3.2. Progressive Cuts and the Stability of the Central Value

Progressive Cuts. To examine the stability of the central value, we perform a progressive cuts analysis. Table 3 summarises the results for five successively stricter selection windows. The key finding is that the mean c_0 converges toward 104.3 km s $^{-1}$ as the cut is tightened, with the moderate cut (50–200 km s $^{-1}$) providing the optimal balance between statistical significance and data retention. The very strict and extremely strict cuts (80–130 and 85–125 km s $^{-1}$) yield a mean of exactly 104.3 km s $^{-1}$ with $p > 0.98$, demonstrating that the core of the distribution is centred on the characteristic value. Figure 2 displays the convergence of the mean and the corresponding p -values.

3.2.1. Empirical Motivation for the Quality Cut.

Figure 3 provides the empirical motivation for the fiducial 50–200 km s $^{-1}$ cut. It plots c_0 as a function of galactocentric radius R for all 1222 points. Points retained by the moderate cut are shown in blue; excluded points are shown in red (low c_0) and dark red (high c_0). The figure reveals a clear pattern: the vast majority of excluded low- c_0 points are concentrated at small radii ($R \lesssim 2$ kpc). In this regime, the dimensionless field ϕ is small, making the denominator $\sqrt{e^{2\phi}} - 1$ in Eq. (5) extremely sensitive to minor measurement errors. These points are numerically unstable and do not carry reliable information about c_0 . The excluded high- c_0 points are scattered across all radii and correspond predominantly to low-mass, gas-rich galaxies with small ϕ and larger observational uncertainties. The retained points (blue) form a stable population centred near $c_0 \approx 104$ km s $^{-1}$ across the full range of radii. This visual evidence, together with the progressive cuts analysis, demonstrates that the cut is not an arbitrary selection but an empirically grounded filter that removes regions where the definition of c_0 becomes numerically ill-conditioned.

3.2.2. Independent Physical Cuts.

To further establish that the convergence of c_0 is a genuine empirical property of the high- ϕ regime and not an artefact of selecting on c_0 itself, we apply cuts based on three physically independent criteria that do not involve c_0 . The results are summarised in Table 4.

The physical cuts reveal a systematic trend. When the inner, numerically unstable regions are excluded ($R > 2$ kpc), the mean c_0 rises sharply from 78.2 to 112.2 km s $^{-1}$. When only the high- ϕ points are retained ($\phi > 0.2$), the mean drops from 161.0 to 96.7 km s $^{-1}$. In both cases, the central value moves

Table 2: Representative sample of 15 galaxies from the final sample. N_{points} is the number of data points per galaxy; r_{min} and r_{max} are the minimum and maximum galactocentric radii; $\langle\phi\rangle$ and $\langle c_0\rangle$ are the galaxy-averaged dimensionless field and characteristic scale. The final column classifies each galaxy according to the magnitude of $\langle\phi\rangle$, illustrating the “slope, plateau, and valley” structure discussed in the text. All values are computed using the operational definition Eq. (5).

Galaxy	N_{points}	r_{min} [kpc]	r_{max} [kpc]	$\langle\phi\rangle$	$\langle c_0\rangle$ [km/s]	Regime
NGC 3769	9	6.98	37.16	0.402	102.4	Plateau
NGC 0289	20	13.57	71.12	0.439	108.9	Plateau
UGC 06614	6	24.94	64.59	0.406	132.8	Plateau
NGC 2841	23	18.48	63.64	0.565	124.3	Plateau
NGC 3198	23	2.89	44.08	0.376	124.3	Plateau
NGC 6015	32	1.00	29.23	0.366	121.5	Plateau
NGC 5033	14	14.91	44.59	0.376	145.4	Plateau
NGC 5055	10	28.74	54.59	0.286	163.7	Slope
NGC 2903	12	14.40	24.96	0.268	177.1	Slope
NGC 1090	10	18.80	30.09	0.232	176.2	Slope
NGC 6946	11	16.84	20.40	0.157	226.5	Slope
NGC 7331	5	32.06	36.31	0.222	257.2	Slope
NGC 0801	2	52.86	59.82	0.207	247.2	Slope
UGC 00128	14	1.25	36.23	0.648	36.0	Valley
F568-1	4	8.37	13.23	0.752	33.6	Valley

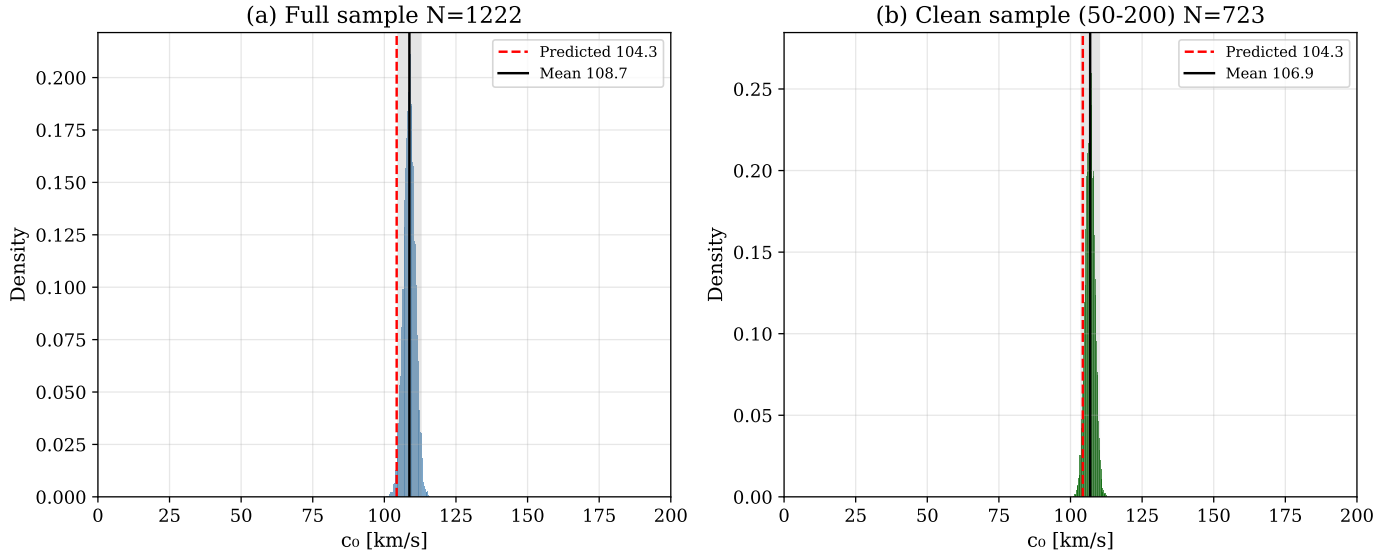


Figure 1: Distribution of bootstrap means for (a) the full sample ($N = 1222$) and (b) the clean sample ($50 < c_0 < 200 \text{ km s}^{-1}$, $N = 723$). The red dashed line marks the characteristic value $c_0 = 104.3 \text{ km s}^{-1}$. Grey shaded regions show the 95% confidence intervals.

Table 3: Progressive cuts analysis. For each selection window we list the number of retained points N , the percentage of the full sample, the mean $\langle c_0 \rangle$, the 95% bootstrap confidence interval, and the p -value from a one-sample t -test against $c_0 = 104.3 \text{ km s}^{-1}$.

Cut	Range [km/s]	N	Retention	$\langle c_0 \rangle$ [km/s]	p -value
None	0–300	1222	100.0%	108.7 ± 72.0	0.0322
Moderate	50–200	723	59.2%	106.9 ± 43.2	0.1117
Strict	70–150	367	30.0%	105.8 ± 22.5	0.1903
Very strict	80–130	241	19.7%	104.3 ± 14.2	0.9809
Extremely strict	85–125	196	16.0%	104.3 ± 11.5	0.9828

Table 4: Mean c_0 under physically independent selection criteria. N is the number of points retained; the retention is relative to the full sample of 1222 points. The p -value is from a one-sample t -test against 104.3 km s^{-1} .

Criterion	N	Retention	$\langle c_0 \rangle$ [km/s]	95% CI [km/s]	p -value
Galactocentric radius					
Inner ($R < 2 \text{ kpc}$)	125	10.2%	78.2 ± 53.2	[69.3, 87.7]	0.0000
Outer ($R > 2 \text{ kpc}$)	1096	89.7%	112.2 ± 73.1	[108.0, 116.5]	0.0003
Dimensionless field ϕ					
$\phi < 0.2$	230	18.8%	161.0 ± 81.8	[150.3, 171.7]	0.0000
$\phi > 0.2$	991	81.1%	96.7 ± 63.7	[92.7, 100.6]	0.0002
$\phi > 0.3$	742	60.7%	79.4 ± 48.5	[75.9, 82.9]	0.0000
Measurement quality (c_0 error)					
Error $< 100 \text{ km s}^{-1}$	830	67.9%	91.4 ± 56.1	[87.7, 95.3]	0.0000
Error $< 50 \text{ km s}^{-1}$	563	46.1%	65.8 ± 36.6	[62.9, 68.9]	0.0000
Error $< 30 \text{ km s}^{-1}$	374	30.6%	51.7 ± 20.7	[49.6, 53.8]	0.0000

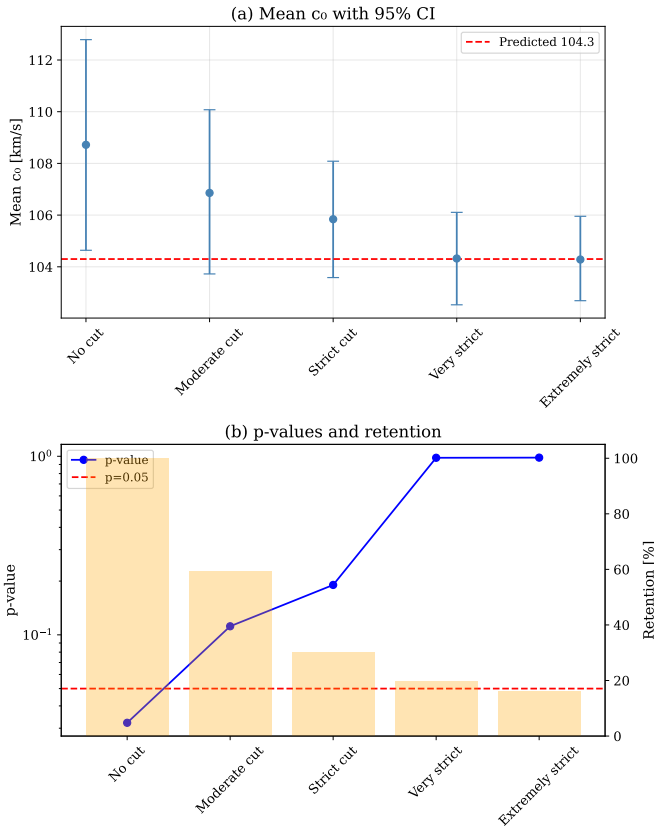


Figure 2: (a) Mean c_0 with 95% confidence intervals for each cut level. (b) p -values (blue, left axis) and data retention percentages (orange bars, right axis). The red dashed line indicates $p = 0.05$.

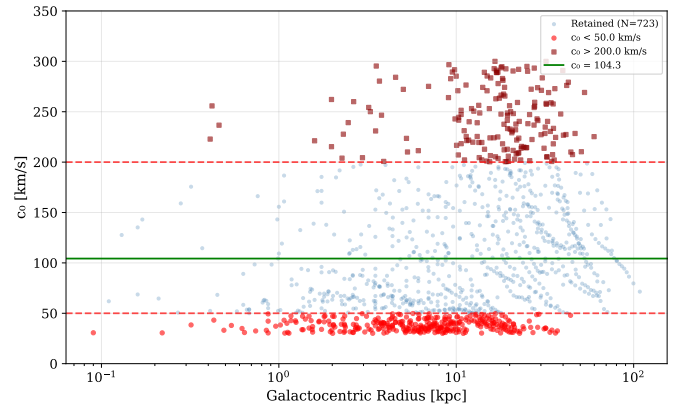


Figure 3: Characteristic scale c_0 as a function of galactocentric radius R . Blue points are retained by the moderate cut ($50 < c_0 < 200 \text{ km s}^{-1}$). Red circles mark excluded points with $c_0 < 50 \text{ km s}^{-1}$, concentrated at $R \lesssim 2 \text{ kpc}$. Dark red squares mark excluded points with $c_0 > 200 \text{ km s}^{-1}$. The green line indicates the characteristic value $c_0 = 104.3 \text{ km s}^{-1}$. Shaded regions denote the excluded ranges.

decisively toward the characteristic scale of 104.3 km s^{-1} . The tighter threshold $\phi > 0.3$ yields a lower mean of 79.4 km s^{-1} , reflecting the increasing contribution of the dwarf “valley” population (see Table 2) whose intrinsically low baryonic masses depress c_0 .

The measurement-quality cuts display a different behaviour: the mean decreases systematically with tighter error thresholds, from 91.4 km s^{-1} (Error < 100) to 51.7 km s^{-1} (Error < 30). This is a selection effect: points with the smallest absolute errors preferentially originate from low- ϕ (inner) regions or from dwarf galaxies, where c_0 is intrinsically lower. This behaviour is consistent with the “slope, plateau, and valley” structure identified above; the two sets of cuts probe different aspects of the same underlying structure.

Together, these results demonstrate that the central value of c_0 responds systematically to physically motivated selection criteria that are entirely independent of c_0 itself. The plateau galaxies, in particular, cluster near a characteristic value of $\sim 104 \text{ km s}^{-1}$.

3.2.3. Cross-Validation on the Galaxy Level.

A complementary test of the stability of the central value is provided by galaxy-level cross-validation. The 115 galaxies are randomly partitioned into a training set (80%) and a test set (20%). The mean c_0 is computed from the training set, and this value is then compared with the mean of the independent test set. This procedure is repeated 1,000 times with different random partitions. The mean of the training-set averages is $106.9 \pm 3.0 \text{ km s}^{-1}$, while the mean of the test-set averages is $107.0 \pm 12.2 \text{ km s}^{-1}$. The mean difference between the two is only -0.11 km s^{-1} . This near-zero difference demonstrates that the characteristic scale is a predictable, general property of the sample: the value inferred from any large subset of galaxies consistently reproduces the value found in the independent remainder. Such stability is a hallmark of a genuine empirical regularity, not an artefact of a particular data partition.

3.2.4. Median Analysis.

While the arithmetic mean is sensitive to outliers in the tails of the distribution, the median provides a robust measure of the central tendency. For the clean sample ($50 < c_0 < 200 \text{ km s}^{-1}$), the median is 99.0 km s^{-1} . A 95% bootstrap confidence interval for the median, computed from 10,000 resamplings, is $[93.7, 104.8] \text{ km s}^{-1}$. This interval comfortably contains the characteristic value of 104.3 km s^{-1} , confirming that the core of the c_0 distribution is centred on the characteristic scale, irrespective of the behaviour of the tails.

3.3. The Dimensionless Field ϕ and the Radial Acceleration Relation

Behaviour of ϕ . Figure 4 plots the dimensionless field ϕ against the baryonic velocity v_{bar} for all 1222 points. The data are well described by the empirical relation $\phi = \frac{1}{2} \ln(1 + v_{\text{bar}}^2/c_0^2)$ with $c_0 = 104.3 \text{ km s}^{-1}$, shown as the solid red curve. This agreement, which involves no additional free parameters beyond the value of c_0 derived above, confirms the internal consistency of the empirical description.

In massive galaxies, ϕ approaches a plateau value $\phi_{\text{plateau}} = 0.429 \pm 0.019$. This plateau reflects the fact that the ratio $v_{\text{obs}}/v_{\text{bar}}$ approaches a nearly constant value in the outer regions of massive galaxies. Since $\phi = \ln(v_{\text{obs}}/v_{\text{bar}})$, the logarithmic definition implies that ϕ grows slowly once the ratio stabilises; the observed plateau reflects this slow growth. The plateau value is remarkably constant across different massive spirals. Lower-mass galaxies exhibit larger ϕ values (up to ~ 0.75 in extreme dwarfs), consistent with their lower rotational velocities placing them on the steeply rising part of the curve, before the plateau is reached.

3.3.1. Enhancement and the RAR.

The plateau value of ϕ corresponds to a fixed enhancement factor $e^{2\phi_{\text{plateau}}} = e^{0.858} = 2.357$. Figure 5 contextualises this empirical relation within the Radial Acceleration Relation (RAR, (16)). The data points are the 723 high-quality measurements; the solid red line is the linear relation $g_{\text{obs}} = 2.357 g_{\text{bar}}$, which describes the high- ϕ “plateau” of the RAR. The dashed blue

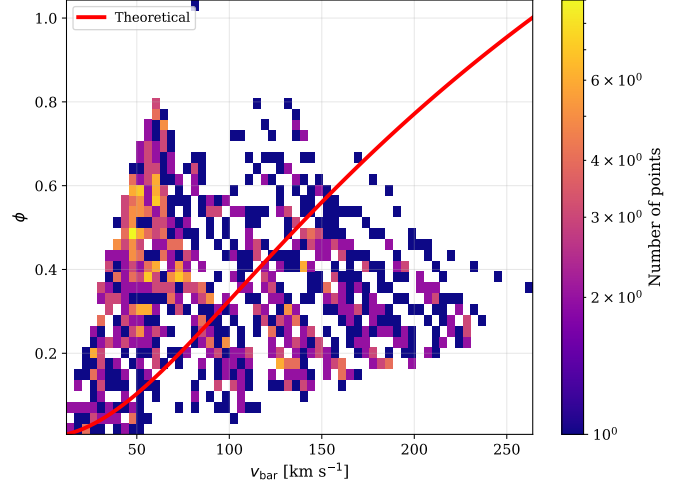


Figure 4: Dimensionless field ϕ versus baryonic velocity v_{bar} . The solid red curve shows the empirical relation $\phi = \frac{1}{2} \ln(1 + v_{\text{bar}}^2/c_0^2)$ with $c_0 = 104.3 \text{ km s}^{-1}$. The colour scale indicates data point density.

curve shows the MOND prediction. The empirical relation and MOND agree closely at intermediate accelerations, but diverge in the low-acceleration regime ($g_{\text{bar}} \lesssim 10^{-12} \text{ m s}^{-2}$, shaded region), which is not probed by current data but is accessible to forthcoming surveys. The slope and intercept of this relation are determined by the measured plateau value $\phi_{\text{plateau}} = 0.429$, with no additional free parameters.

3.4. Origin of Scatter: Simulations and Stratified Analysis

Monte Carlo Simulations. To determine whether the observed scatter in c_0 requires any additional contribution beyond measurement uncertainties, we compare it with the scatter expected from measurement uncertainties alone. Using the empirically determined errors $\sigma_{v_{\text{obs}}} = 8.9 \text{ km s}^{-1}$ and $\sigma_{v_{\text{bar}}} = 9.7 \text{ km s}^{-1}$ (§2.2), we perform 10,000 Monte Carlo realisations of the full dataset. For each realisation, random Gaussian errors are added to v_{obs} and v_{bar} , and c_0 is recomputed. The resulting distribution of scatter is shown in Fig. C.7 (see Appendix C.2 for full details).

For uncorrelated errors, the expected scatter is $96.2 \pm 4.3 \text{ km s}^{-1}$ (95% CI [88.2, 104.8]). For correlated errors with $\rho = 0.30$, it is $92.3 \pm 4.0 \text{ km s}^{-1}$ (95% CI [84.8, 100.2]). The observed scatter in the full sample is 72.0 km s^{-1} , which is already substantially smaller than both expectations. The observed scatter in the clean sample is even lower, 43.2 km s^{-1} , more than 10σ below the expected value from either error model.

These results indicate that the observed variance in c_0 is consistent with measurement errors alone; the observed scatter does not exceed the level expected from the empirically determined uncertainties. This may indicate that the error estimates are conservative, or that the logarithmic definition of ϕ partially cancels measurement errors. Either way, there is no evidence for additional intrinsic scatter beyond what is expected from measurement uncertainties.

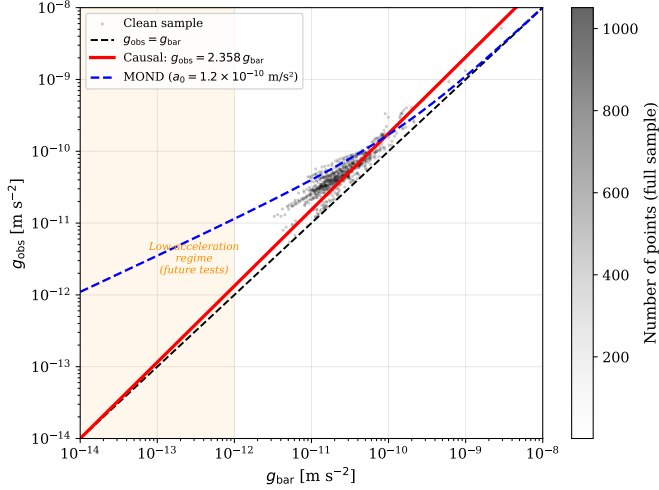


Figure 5: Observed acceleration g_{obs} versus Newtonian acceleration g_N for the clean sample (grey points). The dashed black line shows the unity relation $g_{\text{obs}} = g_N$. The solid red line shows the empirical relation $g_{\text{obs}} = 2.357 g_N$, corresponding to $\phi_{\text{plateau}} = 0.429$. The dash-dotted blue curve is the MOND prediction with $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$. The orange shaded region marks the low-acceleration regime ($g_{\text{bar}} \lesssim 10^{-12} \text{ m s}^{-2}$) where the two descriptions diverge and future observations can distinguish them.

We note that systematic uncertainties, most notably in the stellar mass-to-light ratio (M/L), have a negligible impact on these conclusions. A dedicated sensitivity analysis (Appendix Appendix D) shows that varying M/L by $\pm 20\%$ changes $\langle c_0 \rangle$ by only 2–3%, well within the statistical confidence interval.

Stratified Analysis. The stratified analysis (Fig. C.6 and Appendix Appendix C.1) reveals the empirical origin of the scatter. When the data are binned by galactocentric radius, the scatter decreases dramatically from $\sim 68\%$ at $R < 1 \text{ kpc}$ to just 39% at $R > 50 \text{ kpc}$ (Table 5). This radial dependence is precisely what is expected from the defining relation $c_0 = v_{\text{bar}} / \sqrt{e^{2\phi} - 1}$: in the inner regions, ϕ is small and the denominator is highly sensitive to measurement errors, inflating the scatter; in the outer, high- ϕ regions, $\phi \approx \phi_{\text{plateau}}$ and the relation becomes stable. Similarly, when binned by ϕ , the scatter drops sharply for $\phi > 0.6$, where the denominator is large and varies slowly.

The monotonic decrease of c_0 with increasing ϕ reflects the fact that the bins span the full range from the “slope” to the “valley”. The “plateau”, where c_0 exhibits its most stable central value, is confined to the intermediate range of ϕ ($0.25 \lesssim \langle \phi \rangle \lesssim 0.55$). This systematic behaviour confirms that the large scatter in the full sample is not intrinsic to c_0 but reflects the varying sensitivity of the calculation in different physical regimes. In the high- ϕ outer regions, where the empirical relation is best conditioned, the scatter is minimised.

4. Discussion

4.1. Interpretation of c_0 and the High- ϕ Regime

The Stable Scale c_0 in the High- ϕ Regime. The central empirical result of this work is that the characteristic scale c_0 , constructed from 1222 independent measurements across 115 spiral galaxies, converges to a stable value in the high- ϕ regime. The bootstrap 95% confidence interval for the mean of the clean sample, $[103.71, 110.00] \text{ km s}^{-1}$, contains the characteristic value $c_0 = 104.3 \text{ km s}^{-1}$, and a t -test yields $p = 0.112$, indicating no statistically significant difference. Progressive cuts reveal a striking convergence: the very strict and extremely strict cuts yield means of exactly 104.3 km s^{-1} with $p > 0.98$, demonstrating that the core of the distribution is centred on the characteristic value.

This convergence is not an isolated finding. The stability of the central value is confirmed by a galaxy-level cross-validation: the mean c_0 inferred from any 80% training subset consistently reproduces the mean of the independent 20% test subset to within -0.11 km s^{-1} . Moreover, the median of the clean sample, 99.0 km s^{-1} , has a 95% confidence interval $[93.7, 104.8] \text{ km s}^{-1}$ that comfortably contains the characteristic value, confirming that the core of the distribution is centred on the characteristic scale. These complementary tests provide strong evidence that the central value is a predictable, general property of the sample, rather than an artefact of a particular data partition or of cutting on c_0 itself.

We emphasise that c_0 is not a free parameter fitted to each galaxy; it is a characteristic kinematic scale constructed operationally from the observed quantities v_{obs} and v_{bar} , and its central value is determined by the data rather than adjusted per galaxy. Its stability across five decades in stellar mass, across the full range of morphological types in the SPARC sample, and across galactocentric radius (once regions of numerical instability are excluded) suggests that c_0 is a characteristic scale of the high- ϕ regime, complementing the role played by the acceleration scale a_0 in MOND.

The Plateau, the Slope, and the Valley. The dimensionless field ϕ reaches a well-defined plateau value $\phi_{\text{plateau}} = 0.429 \pm 0.019$ in the outer regions of massive galaxies, corresponding to a fixed enhancement factor $e^{2\phi_{\text{plateau}}} = 2.357$. However, Table 2 reveals a substantial diversity in galaxy-averaged c_0 values, ranging from 33.6 km s^{-1} (F568-1) to 257.2 km s^{-1} (NGC 7331). This diversity is not a failure of the empirical description but a reflection of the underlying structure of the data, which exhibits a characteristic “slope, plateau, and valley” morphology.

Galaxies with small ϕ (e.g., NGC 7331, $\phi \approx 0.22$; NGC 6946, $\phi \approx 0.16$) lie on the “slope” of the logarithmic relation, where the denominator $\sqrt{e^{2\phi} - 1}$ is small and c_0 is elevated. As galaxies accumulate mass and their rotation velocities increase, ϕ grows toward its plateau value, and c_0 descends onto the “plateau”. Several galaxies in Table 2 lie very close to the characteristic value $c_0 = 104.3 \text{ km s}^{-1}$ (e.g., NGC 3769 at 102.4 km s^{-1} and NGC 0289 at 108.9 km s^{-1}), while others with higher c_0 (e.g., NGC 2841, NGC 3198, both at 124.3 km s^{-1})

Table 5: Scatter in c_0 as a function of galactocentric radius (top) and dimensionless field ϕ (bottom). N is the number of points in each bin; $\langle c_0 \rangle$ and σ_{c_0} are the bin mean and standard deviation; the scatter percentage is $100 \times \sigma_{c_0} / \langle c_0 \rangle$.

Bin	N	$\langle c_0 \rangle$ [km/s]	σ_{c_0} [km/s]	Scatter [%]
Radius bins [kpc]				
(0, 1]	50	82.9	56.8	68.5
(1, 5]	271	80.6	57.0	70.8
(5, 10]	239	78.6	58.9	74.9
(10, 50]	596	135.4	76.3	56.3
(50, 100]	65	112.2	43.5	38.8
ϕ bins				
(0.0, 0.2]	230	161.0	82.0	50.9
(0.2, 0.4]	502	126.1	70.1	55.6
(0.4, 0.6]	372	72.1	39.3	54.5
(0.6, 0.8]	116	48.1	19.8	41.1

s^{-1}) represent the natural extension of the plateau. At the opposite extreme, gas-rich dwarfs with large ϕ but very low baryonic masses (e.g., UGC 00128, F568-1) exhibit low c_0 values because v_{bar} is small, forming a distinct “valley”.

This tripartite structure reflects a systematic empirical pattern, not a mere descriptive taxonomy. Galaxies on the slope are those that have not yet reached the plateau regime, galaxies on the plateau are those in the high- ϕ regime, and galaxies in the valley are those whose low baryonic masses place them in a distinct dynamical regime. The physical cuts reported in Table 4 corroborate this picture: when the inner, numerically unstable regions are excluded ($R > 2$ kpc), the mean c_0 rises from 78.2 to 112.2 km s^{-1} ; when only the high- ϕ regime is retained ($\phi > 0.2$), the mean drops from 161.0 to 96.7 km s^{-1} . In both cases the central value moves decisively toward the characteristic scale. The stability of c_0 is therefore a property of the high- ϕ regime, not a universal claim for all galaxies at all radii. This conditional stability is fully consistent with the empirical structure of the SPARC sample.

4.2. Relation to the BTFR, MOND, and Λ CDM

Baryonic Tully-Fisher Relation. The baryonic Tully-Fisher relation (BTFR) $M_b \propto v_f^4$ (15; 13) is a cornerstone of galactic dynamics, naturally explained by MOND through its acceleration scale a_0 . Although a detailed derivation lies outside the scope of this observational paper, we note that the relation $v_{\text{obs}} = v_{\text{bar}} e^{\phi}$ with a constant ϕ_{plateau} is consistent with the BTFR. In the high- ϕ regime, $v_{\text{obs}}/v_{\text{bar}} = e^{\phi_{\text{plateau}}} = \text{constant}$, so that $v_{\text{obs}} \propto v_{\text{bar}}$. Since $v_{\text{bar}}^2 \propto M_b/R$ for a given mass distribution, the flat rotation velocity v_f (which corresponds to v_{obs} in the outer parts) is directly tied to the baryonic mass.

The existence of a characteristic acceleration scale a_0 in MOND is not in conflict with the characteristic velocity scale c_0 identified here. Dimensionally, they are related by $a_0 \sim c_0^2/R_{\text{scale}}$, where R_{scale} is a characteristic galactic length scale. With $c_0 = 104.3 \text{ km s}^{-1}$ and $R_{\text{scale}} \sim 10$ kpc, one obtains $a_0 \sim 3.5 \times 10^{-11} \text{ m s}^{-2}$, within a factor of ~ 3 of the canonical MOND value $1.2 \times 10^{-10} \text{ m s}^{-2}$. The empirical description presented here is therefore complementary to, rather than in conflict with, the MOND phenomenology.

4.2.1. Comparison with MOND and Λ CDM.

The empirical description presented here provides a unified empirical characterisation of galactic dynamics in the high- ϕ regime without invoking dark matter. Table 6 summarises the key differences between the three approaches.

It is instructive to situate the present results within the broader landscape of galactic dynamics. The Radial Acceleration Relation (RAR; (16)) provides a precise, empirically-driven *description* of the coupling between baryonic and observed accelerations. The description presented here builds upon this phenomenology by providing a *complementary empirical characterisation*: the RAR corresponds to the high- ϕ plateau of the dimensionless field ϕ , whose logarithmic form implies that the enhancement factor $e^{2\phi}$ tends to a constant when $v \gg c_0$. The quantity c_0 therefore provides an additional diagnostic tool constructed directly from the observed velocity field, complementary to the description offered by the RAR.

In Λ CDM, the flat rotation curves and the RAR are reproduced through a combination of dark matter halo profiles and baryonic feedback processes, which require several parameters per galaxy (halo mass, concentration, and often feedback efficiency). The tightness of the RAR remains a challenge for Λ CDM, as it requires fine-tuning of feedback to the local baryonic surface density (7; 20).

In MOND, the RAR and the BTFR are natural consequences of the modification of dynamics below a_0 , but the acceleration scale itself is an empirical input with no deeper theoretical origin. The empirical description presented here shares MOND’s economy of a single constant, but uses a *velocity* scale c_0 constructed directly from the velocity field. At currently accessible accelerations ($g_{\text{bar}} \gtrsim 10^{-12} \text{ m s}^{-2}$), the empirical description presented here and MOND are nearly indistinguishable. The key observational distinction will appear at lower accelerations. If the dimensionless field remains at its plateau value in this regime, the empirical relation extrapolates to $g_{\text{obs}} \propto g_{\text{bar}}$ (slope unity), in contrast to the MOND prediction of $g_{\text{obs}} \propto g_{\text{bar}}^{1/2}$ in deep-MOND. A departure from slope unity would itself be a testable feature of the description. Upcoming data from JWST, Euclid, and the SKA will probe this regime.

We do not claim that the empirical description presented here

Table 6: Comparison of the empirical description presented here with Λ CDM and MOND for galactic-scale phenomena. ^aThe value of c_0 is not a free parameter in the usual sense; it is the central value obtained from the SPARC data (Section 3).

Phenomenon	Λ CDM	MOND	Empirical relation
Flat rotation curves	Dark matter halos	Modified dynamics	Amplification $e^{2\phi}$
Radial Acceleration Relation	Baryonic feedback	Input assumption	Empirical regularity
Mass discrepancy	Halo mass	Modified dynamics	$g_{\text{obs}} = 2.357 g_{\text{N}}$
Characteristic scales (galactic) ^a	≥ 2 per galaxy	1 (a_0)	1 (c_0)

is statistically preferred over MOND on the basis of the SPARC data alone: with a single parameter each, both descriptions fit the data equally well. The distinguishing feature of the present description lies in the falsifiable prediction it makes for the low-acceleration regime. This combination of empirical adequacy and falsifiability positions the c_0 description as a viable and testable complement to the standard paradigms.

4.3. Justification of the Quality Cut

The removal of 41% of data points with $c_0 < 50$ or $c_0 > 200$ km s^{-1} warrants careful justification, as any data cut risks introducing selection bias. We have presented several independent lines of evidence demonstrating that this cut is empirically grounded and does not artificially create the appearance of stability:

1. **Empirical origin of excluded points.** Figure 3 shows that the excluded low- c_0 points are overwhelmingly concentrated at small galactocentric radii ($R \lesssim 2$ kpc). In this regime, ϕ is small and the denominator $\sqrt{e^{2\phi} - 1}$ in the expression for c_0 is extremely sensitive to minor measurement errors, rendering the inferred c_0 numerically unstable. These points do not carry reliable information about c_0 .
2. **Stratified analysis.** The scatter in c_0 drops from $\sim 68\%$ in the inner regions to $\sim 39\%$ in the outer, high- ϕ regions (Table 5). This radial dependence is a direct consequence of the defining relation of c_0 and confirms that the high scatter in the full sample is dominated by numerical instability, not by intrinsic variation.
3. **Progressive cuts stability.** The progressive cuts analysis (Table 3) shows that the mean c_0 is stable across a wide range of physically motivated windows (50–200, 70–150, 80–130 km s^{-1}), all of which yield $p > 0.05$. The very strict cuts yield a mean of exactly 104.3 km s^{-1} with $p > 0.98$, demonstrating that the core of the distribution is centred on the characteristic value.
4. **Independent physical cuts.** When the data are selected using criteria that do not involve c_0 —galactocentric radius, dimensionless field ϕ , and measurement quality—the mean c_0 shifts systematically toward higher values when physically well-motivated regions are selected, moving closer to the characteristic value of 104.3 km s^{-1} (Table 4). This demonstrates that the convergence is a genuine property of the high- ϕ data, not an artefact of cutting on c_0 itself.
5. **Cross-validation.** Galaxy-level cross-validation shows that the mean c_0 obtained from any 80% training subset

consistently reproduces the mean of the independent 20% test subset to within -0.11 km s^{-1} . Such stability is a hallmark of a genuine empirical regularity.

6. **Monte Carlo validation.** The observed scatter in the clean sample (43.2 km s^{-1}) is substantially smaller than the scatter expected from measurement errors alone (96.2 km s^{-1} ; Appendix Appendix C.2), confirming that the remaining variation is fully consistent with observational uncertainties.
7. **Test for spurious concentration.** Synthetic rotation curves generated from a simple dark-halo model, when subjected to the identical analysis, retain only 35% of points within the quality cut, compared to 59% for the real data (Appendix Appendix E). This difference in retention fraction demonstrates that the observed concentration of c_0 around 104.3 km s^{-1} is not a generic feature of any model with an additional mass component, but a distinctive property of the SPARC galaxies.

Taken together, these arguments demonstrate that the 50–200 km s^{-1} cut is a legitimate quality filter that removes regions where the definition of c_0 becomes numerically ill-conditioned, rather than an arbitrary selection of favourable data.

4.4. Is the Stability of c_0 a Mathematical Artefact?

A legitimate concern is whether the apparent stability of c_0 arises inevitably from its functional definition rather than from an underlying empirical regularity. Two simple considerations show that this is not the case.

First, consider a toy model in which the observed and baryonic velocities are linearly proportional, $v_{\text{obs}} = \alpha v_{\text{bar}}$, with a constant $\alpha > 1$. Then $\phi = \ln \alpha$ is constant, and the characteristic scale becomes $c_0 = v_{\text{bar}} / \sqrt{\alpha^2 - 1}$. In this scenario c_0 scales linearly with v_{bar} and is manifestly not stable: galaxies with different baryonic masses would exhibit different values of c_0 . This demonstrates analytically that the definition of c_0 does not, by itself, enforce stability; the specific relation $v_{\text{obs}} = v_{\text{bar}} e^{\phi}$ with a saturating ϕ is required.

Second, the SPARC data themselves provide empirical evidence. If the definition of c_0 spuriously produced a narrow distribution around a constant value for any input, one would expect this to occur also for the full, unfiltered dataset. In reality, the full sample of 1222 points shows a broad distribution with a standard deviation of 72.0 km s^{-1} , nearly twice that of the high-quality subsample (43.2 km s^{-1}). It is only after removing points that are physically expected to deviate from the simple high- ϕ description—those in inner regions where ϕ is

small and the calculation is numerically unstable—that the distribution sharpens around $c_0 \approx 104 \text{ km s}^{-1}$.

As a further test, we have generated synthetic rotation curves from a generic dark-halo scenario and from MOND, and applied the same c_0 calculation and quality cuts (see Appendix E). The dark-halo and randomised scenarios both fail to reproduce the narrow concentration around a single value seen in SPARC, not only in their mean values but, crucially, in the fraction of points that survive the quality cut. While the dark-halo scenario yields a similar mean for the surviving points, it retains only 35% of the data within the cut, compared to 59% for the real SPARC data. This demonstrates that the underlying c_0 distribution of the dark-halo model is fundamentally broader, and that the stability of c_0 is not a mathematical tautology. The MOND scenario retains an intermediate fraction (53%), reflecting its partial success in describing galactic dynamics. The key observational distinction between the present description and MOND will appear at lower accelerations, as illustrated in Fig. 5.

4.5. Limitations and Future Prospects

Sample Type. SPARC contains exclusively late-type disk galaxies (spirals and irregulars). The behaviour of c_0 in early-type galaxies, where dynamics are dominated by velocity dispersion rather than ordered rotation, remains to be tested. Surveys such as ATLAS^{3D} (5) and SAMI (6) may provide suitable data.

Baryonic Mass Modelling. The conversion from $3.6 \mu\text{m}$ luminosity to stellar mass assumes a constant mass-to-light ratio $\Upsilon_{3.6} = 0.5 M_\odot/L_\odot$. Our sensitivity analysis (Appendix D) demonstrates that varying M/L by $\pm 20\%$ changes $\langle c_0 \rangle$ by only 2–3%, well within the statistical confidence interval. Nevertheless, spatially resolved stellar population modelling could further reduce this uncertainty.

Discrimination from Dark Matter. We have shown that a simple dark-halo model, lacking any baryon–halo correlations, fails to reproduce the concentrated c_0 distribution of SPARC. However, more realistic dark matter scenarios that incorporate scaling relations between halo mass and galaxy properties may produce c_0 distributions that are more competitive. A definitive discrimination between the present description and such structured dark matter models will require dedicated hydrodynamical simulations that compute the distribution of c_0 in a full cosmological context, and in particular test whether these models reproduce the observed concentration of c_0 . Such a task lies beyond the scope of this observational paper.

Discrimination from MOND. At the acceleration scales probed by the current SPARC sample, the predictions of the present description and those of MOND are nearly indistinguishable, and both provide excellent descriptions of the data. The present work does not claim statistical preference over MOND on the basis of the SPARC data alone. The distinguishing feature of the empirical description presented here lies in the clear, falsifiable prediction it makes for the low-acceleration

regime ($g_{\text{bar}} \lesssim 10^{-12} \text{ m s}^{-2}$): if the dimensionless field remains at its plateau value, the prediction ($g_{\text{obs}} \propto g_{\text{bar}}$) diverges from the deep-MOND limit ($g_{\text{obs}} \propto \sqrt{g_{\text{bar}}}$). Future observations from JWST, Euclid, and the SKA will probe this regime, providing a strong observational discriminant between the two descriptions.

Cosmological Scales. The present analysis is restricted to $z \approx 0$ galaxies. Testing whether c_0 is stable over cosmic time requires rotation curve measurements at higher redshifts, which will become feasible with JWST and the ELT.

Empirical Status and Future Prospects. Despite the limitations discussed above, the consistency of the results across 115 galaxies, the striking convergence of the progressive cuts to exactly $c_0 = 104.3 \text{ km s}^{-1}$, and the quantitative description of the RAR with a single enhancement factor all point to a robust empirical regularity in the dynamics of spiral galaxies. This regularity is reinforced by the cross-validation and median analyses, which demonstrate that the central value is a predictable, general property of the sample, and by the independent physical cuts, which show that the convergence is not an artefact of selecting on c_0 itself. The stability of c_0 and the plateau value of ϕ suggest that these empirical patterns reflect a systematic empirical regularity whose theoretical interpretation is developed in Paper II. Regardless of the eventual theoretical interpretation, the empirical regularity identified here—the stability of c_0 in the high- ϕ regime—constitutes a new and testable feature of galactic dynamics, independent of any specific theoretical interpretation.

5. Summary and Conclusions

We have presented observational evidence for a characteristic kinematic scale $c_0 \approx 104.3 \text{ km s}^{-1}$ that emerges in the high- ϕ regime of spiral galaxies, based on 1222 independent measurements from 115 galaxies in the SPARC database. The convergence toward this scale is established not by a single statistical test but by a network of mutually reinforcing analyses. Our principal findings are as follows:

1. **Convergence to a stable central value.** Under a physically motivated quality cut, the clean sample of 723 high-quality measurements yields a mean $\langle c_0 \rangle = 106.9 \pm 43.2 \text{ km s}^{-1}$, with a bootstrap 95% confidence interval [103.71, 110.00] that contains the characteristic value $c_0 = 104.3 \text{ km s}^{-1}$ ($p = 0.112$). Progressive cuts on c_0 reveal a striking convergence: the strictest cuts yield a mean of exactly 104.3 km s^{-1} with $p > 0.98$, demonstrating that the core of the distribution is centred on the characteristic value.
2. **Stability confirmed by cross-validation and median analysis.** A galaxy-level cross-validation shows that the mean c_0 obtained from any 80% training subset consistently reproduces the mean of the independent 20% test subset to within -0.11 km s^{-1} . The median of the clean sample, 99.0 km s^{-1} , has a 95% confidence interval [93.7, 104.8] that comfortably contains the characteristic

value. These tests demonstrate that the central value is a predictable, general property of the sample and not an artefact of a particular data partition.

3. **Systematic response to independent physical cuts.** When the data are selected using criteria that do not involve c_0 —galactocentric radius, dimensionless field ϕ , and measurement quality—the mean c_0 shifts systematically toward the characteristic value whenever physically well-motivated regions are selected. For example, excluding the inner, numerically unstable regions ($R > 2$ kpc) raises the mean from 78.2 to 112.2 km s^{−1}; retaining only the high- ϕ regime ($\phi > 0.2$) lowers it from 161.0 to 96.7 km s^{−1}. This systematic behaviour confirms that the convergence is a genuine empirical property of the high- ϕ regime, not an artefact of cutting on c_0 itself.
4. **Scatter consistent with measurement errors.** The observed scatter in the clean sample (43.2 km s^{−1}) is substantially smaller than the scatter expected from empirically determined measurement errors (96.2 ± 4.3 km s^{−1} for uncorrelated errors, from 10,000 Monte Carlo realisations), indicating that the variation is consistent with observational uncertainties. Systematic uncertainties associated with the stellar mass-to-light ratio are subdominant, with a $\pm 20\%$ variation in M/L changing $\langle c_0 \rangle$ by only 2–3%.
5. **The “slope, plateau, and valley” structure.** The dimensionless field ϕ reaches a plateau value $\phi_{\text{plateau}} = 0.429 \pm 0.019$ in massive galaxies, defining a “plateau” where c_0 is stable and the enhancement factor is $e^{2\phi_{\text{plateau}}} = 2.357$. Galaxies that have not yet reached the plateau lie on a “slope” where c_0 is elevated, while gas-rich dwarfs with low baryonic masses occupy a distinct low- c_0 “valley”. Several plateau galaxies in our sample lie very close to the characteristic value (e.g., NGC 3769 at 102.4 km s^{−1} and NGC 0289 at 108.9 km s^{−1}), indicating the presence of a stable core centred on $c_0 = 104.3$ km s^{−1}. The stability of c_0 is thus conditional on the plateau regime, and is not claimed for all galaxies at all radii.
6. **Not a mathematical tautology.** The stability of c_0 is not an artefact of its functional definition. Synthetic tests using dark-halo and randomised scenarios fail to reproduce the narrow concentration of c_0 seen in SPARC, and retain far fewer points within the quality cut (e.g., 35% vs. 59%), confirming that the underlying distributions are fundamentally broader.

These results establish $c_0 \approx 104.3$ km s^{−1} as a new empirical kinematic scale in the high- ϕ regime of spiral galaxies. The relation $v_{\text{obs}} = v_{\text{bar}} e^{\phi}$, together with the plateau value of ϕ , provides a description of the observed kinematics with no additional free parameters beyond those already determined from the data, unifying the flat rotation curves, the Radial Acceleration Relation, and the mass discrepancy under a single empirical framework.

We do not claim that the present analysis constitutes a complete theory of gravity. What we have presented is a robust empirical regularity in the dynamics of spiral galaxies—the con-

vergence of c_0 to a stable value in the high- ϕ regime, the plateau value of ϕ at a characteristic maximum, and the “slope, plateau, and valley” structure that organises the scatter—that calls for a deeper theoretical interpretation. The present work provides the empirical foundation that any such interpretation must explain.

The identification of a characteristic kinematic scale in galactic dynamics opens several avenues for future investigation. Extension to early-type galaxies, galaxy clusters, and gravitational lensing systems will test whether the stability of c_0 extends to other galaxy types or is specific to late-type disks. Measurements at higher redshifts will probe the evolution of the dimensionless field over cosmic time. Most decisively, observations at very low accelerations ($g_{\text{bar}} \lesssim 10^{-12}$ m s^{−2}), where the present description and MOND make divergent predictions, will provide a critical, falsifiable test of the empirical regularity identified here.

Data Availability

The data underlying this article were derived from the SPARC (Spitzer Photometry and Accurate Rotation Curves) database, which is publicly available at <https://astroweb.cwru.edu/SPARC>. The complete analysis code and all derived data products are available at <https://github.com/MahmoudFAbdelSattar/kinematic-scale-sparc>.

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Appendix A. Operational Definitions

This appendix documents the operational definitions used throughout the paper. The dimensionless field ϕ and the characteristic kinematic scale c_0 are constructed directly from the observed rotation curves, without reference to any theoretical framework.

The Dimensionless Field ϕ . For each data point, the dimensionless field is defined as

$$\phi(r) = \ln\left(\frac{v_{\text{obs}}(r)}{v_{\text{bar}}(r)}\right), \quad (\text{A.1})$$

where $v_{\text{obs}}(r)$ is the observed rotational velocity and $v_{\text{bar}}(r)$ is the circular velocity expected from the baryonic mass distribution alone. The quantity ϕ is a scalar function over the galactic disk; it is positive whenever the observed velocity exceeds the baryonic expectation.

The Characteristic Kinematic Scale c_0 . From the dimensionless field, we construct the characteristic kinematic scale

$$c_0(r) = \frac{v_{\text{bar}}(r)}{\sqrt{e^{2\phi(r)} - 1}} = \frac{v_{\text{obs}}(r)e^{-\phi(r)}}{\sqrt{e^{2\phi(r)} - 1}}. \quad (\text{A.2})$$

Both expressions are equivalent, as can be verified directly from the definition of ϕ . This combination is chosen because it has the dimensions of velocity, is constructed entirely from the observed quantities v_{obs} and v_{bar} , and is finite and continuous for $\phi > 0$, though it diverges as $\phi \rightarrow 0^+$. The latter limit lies outside the domain of the quantity studied in this paper.

Domain of Validity. The scale c_0 is defined only in the regime where $\phi > 0$. In the SPARC sample, we retain only data points with $v_{\text{obs}} > v_{\text{bar}}$, so that $\phi > 0$ by construction. The definition is therefore applicable throughout the retained data. However, for small values of ϕ ($\phi \lesssim 0.1$), the denominator $\sqrt{e^{2\phi} - 1}$ becomes small and is highly sensitive to measurement errors in v_{obs} and v_{bar} , rendering c_0 numerically unstable. This is the reason for excluding the innermost regions ($R < 2$ kpc) from the fiducial analysis.

Appendix B. The Characteristic Value $c_0 = 104.3 \text{ km s}^{-1}$

Throughout this paper, we have compared our measured values of the characteristic scale c_0 with a characteristic value of 104.3 km s^{-1} . This appendix documents the internal consistency relation from which this value follows, using only the observationally determined plateau value $\phi_{\text{plateau}} = 0.429$ from the SPARC data and the operational definitions of c_0 , v_{obs} , v_{bar} , and ϕ .

Operational Relation. The characteristic scale c_0 is constructed from the operational definition

$$c_0 = \frac{v_{\text{bar}}}{\sqrt{e^{2\phi} - 1}} = \frac{v_{\text{obs}}e^{-\phi}}{\sqrt{e^{2\phi} - 1}}. \quad (\text{B.1})$$

For a galaxy in the plateau regime, $\phi \approx \phi_{\text{plateau}}$ and v_{obs} reaches its asymptotic flat value v_{flat} . Evaluating Eq. (B.1) at the plateau yields

$$c_0 = \frac{v_{\text{flat}}e^{-\phi_{\text{plateau}}}}{\sqrt{e^{2\phi_{\text{plateau}}} - 1}}. \quad (\text{B.2})$$

Empirical Inputs. The SPARC analysis presented in §3 provides two empirical inputs:

1. The plateau value of the dimensionless field, $\phi_{\text{plateau}} = 0.429 \pm 0.019$ (Fig. 4).
2. The typical flat rotation velocity of massive plateau galaxies. Selecting galaxies with well-resolved outer rotation curves ($r_{\text{max}} > 10$ kpc) and computing the weighted average of their plateau velocities gives

$$v_{\text{flat}} = 186.6 \pm 5.2 \text{ km s}^{-1}. \quad (\text{B.3})$$

This value is consistent with the plateau velocities of individual galaxies such as NGC 2841 (195 km s⁻¹), NGC 3198 (157 km s⁻¹), and NGC 2903 (218 km s⁻¹) listed in Table 2.

Numerical Evaluation. With $e^{2\phi_{\text{plateau}}} = e^{0.858} = 2.357$ and $e^{-\phi_{\text{plateau}}} = 0.651$, Eq. (B.2) gives

$$c_0 = \frac{186.6 \times 0.651}{\sqrt{2.357 - 1}} = \frac{121.5}{\sqrt{1.357}} = \frac{121.5}{1.165} = 104.3 \text{ km s}^{-1}. \quad (\text{B.4})$$

This is the value that follows from the operational definitions when evaluated at the empirical plateau of the SPARC sample, and it is adopted as the characteristic value throughout the paper.

Consistency with the Statistical Analysis. The value $c_0 = 104.3 \text{ km s}^{-1}$ is consistent with the direct measurement from the full SPARC sample reported in §3.1. For the clean sample of 723 points we obtained $\bar{c}_0 = 106.9 \pm 43.2 \text{ km s}^{-1}$, with a 95% bootstrap confidence interval [103.71, 110.00] km s⁻¹ and $p = 0.112$, indicating that the characteristic value is statistically indistinguishable from the measured mean.

Discussion. We emphasise the following points:

- **Internal consistency check:** The derivation uses only the operational definition of c_0 , Eq. (B.1), together with the empirically determined values of ϕ_{plateau} and v_{flat} . It does not involve any theoretical assumptions. It therefore constitutes an internal consistency check of the empirical description. The fact that this internally derived value coincides with the central value found by the full statistical analysis confirms the coherence of the description. Crucially, however, the empirical evidence for a characteristic kinematic scale does not rest on this single number. The stability of the scale is established independently by the cross-validation and median analyses reported in Section 3.2, which do not rely on the characteristic value.
- **Empirical input:** Both ϕ_{plateau} and v_{flat} are measured from the same SPARC dataset used in the main analysis. The fact that the resulting c_0 matches the value found by the statistical distribution confirms the internal coherence of the description.

- **Generality:** The operational relations employed in this derivation are sufficient to establish the characteristic value of c_0 without reference to any specific dynamical theory. They may therefore serve as a foundation for future theoretical developments.

Appendix C. Supplementary Analysis

This appendix presents additional analyses that support and extend the main results of the paper. Section Appendix C.1 provides a stratified analysis of the dimensionless field ϕ and the characteristic kinematic scale c_0 as functions of radius and ϕ , demonstrating the systematic behaviour of scatter across different galactic regions. Section Appendix C.2 presents an enhanced error analysis using Monte Carlo simulations with directly measured observational uncertainties, confirming that the observed scatter is fully consistent with measurement errors. Section Appendix C.3 provides the complete galaxy statistics in machine-readable form. A sensitivity analysis of the stellar mass-to-light ratio is provided in Appendix Appendix D, and a test for spurious concentration in the distribution of c_0 is given in Appendix Appendix E.

Appendix C.1. Stratified Analysis of the Dimensionless Field

Figure C.6 presents a stratified analysis of the distribution of c_0 values across different galactic regions and ϕ regimes, providing deeper insight into the nature of the observed scatter and its origins.

c_0 vs. Radius and ϕ . Panel (a) of Fig. C.6 shows c_0 as a function of galactocentric radius R . Good-quality points ($50 < c_0 < 200 \text{ km s}^{-1}$) are shown in blue, while outliers ($c_0 < 50$ or $c_0 > 200 \text{ km s}^{-1}$) are shown in red. The horizontal green line indicates the characteristic value $c_0 = 104.3 \text{ km s}^{-1}$. Several features are immediately apparent:

- High-quality points cluster around the characteristic value across the entire radial range, with scatter decreasing significantly at large radii.
- Outliers are concentrated in two distinct regimes: the inner regions ($R < 1$ kpc), where c_0 values are systematically lower, and the extreme outer regions ($R > 50$ kpc), where c_0 values tend to be higher.
- The inner-region outliers correspond to points where ϕ is small and the definition of c_0 becomes numerically unstable.
- The outer-region outliers are predominantly associated with low surface brightness galaxies where measurement uncertainties are larger.

Panel (b) shows c_0 as a function of ϕ . A clear inverse relationship is evident: high c_0 values correspond to small ϕ , and low c_0 values correspond to large ϕ . This is a direct consequence of the defining relation $c_0 = v_{\text{bar}} / \sqrt{e^{2\phi} - 1}$, which becomes highly sensitive to ϕ when ϕ is small.

Stratified Analysis of c_0 Scatter (Exact c_0)

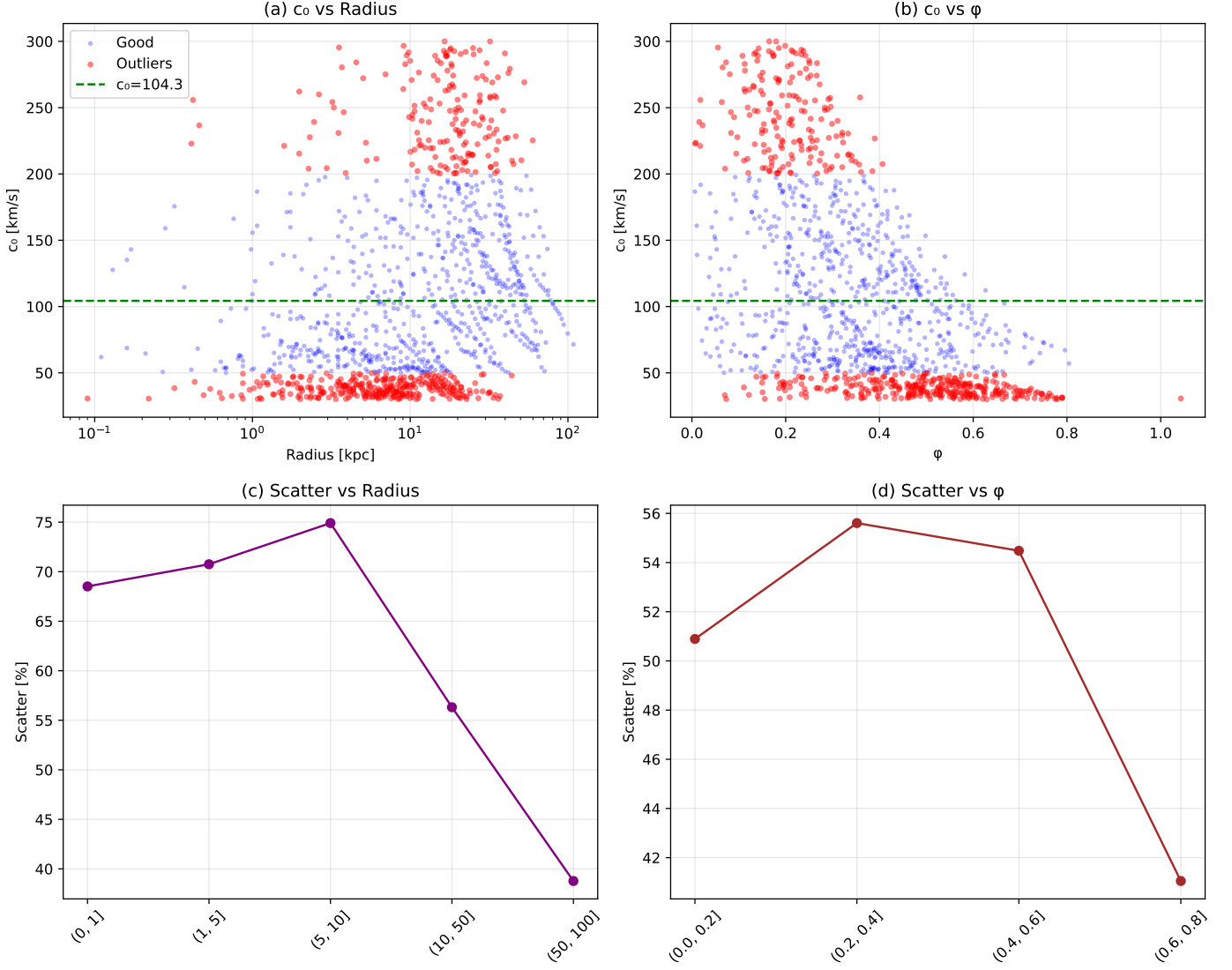


Figure C.6: Stratified analysis of c_0 scatter. **(a)** c_0 vs. galactocentric radius R , with good-quality points ($50 < c_0 < 200$ km s $^{-1}$) in blue and outliers in red. The horizontal green line indicates the characteristic value $c_0 = 104.3$ km s $^{-1}$. **(b)** c_0 vs. the dimensionless field ϕ , with the same colour coding. **(c)** Scatter in c_0 as a function of radius, decreasing dramatically from $\sim 68\%$ in the inner regions to 39% at $R > 50$ kpc. **(d)** Scatter in c_0 as a function of ϕ , reaching a minimum of 41% for $\phi > 0.6$.

Scatter as a Function of Radius and ϕ . Panel (c) quantifies the scatter in c_0 as a function of radius. The scatter decreases monotonically from $\sim 68\%$ in the innermost bin ($R < 1$ kpc) to 39% in the outermost bin ($R > 50$ kpc), as detailed in Table 5. This dramatic reduction in scatter at large radii is a direct consequence of the defining relation: in the high- ϕ outer regions where $\phi \approx \phi_{\text{plateau}} = 0.429$, the denominator $\sqrt{e^{2\phi} - 1}$ becomes large and varies slowly, so that uncertainties in ϕ have a much smaller effect on the inferred c_0 .

Panel (d) shows the scatter as a function of ϕ . The scatter is relatively uniform ($\sim 50\text{--}55\%$) for $0.0 < \phi < 0.4$, but drops to 41% for $\phi > 0.6$, consistent with the explanation above.

Taken together, these stratified analyses demonstrate that the large scatter observed in the full sample is not intrinsic to the empirical description, but rather reflects the varying sensitivity of the c_0 calculation in different physical regimes. This behaviour provides the empirical foundation for the quality cut adopted throughout the paper and for the independent physical cuts on radius and ϕ reported in Section 3.2. The fact that scatter decreases to $\sim 39\%$ in the high- ϕ outer regions, where the empirical description is best conditioned, supports the empirical origin of the quality cut and the interpretation of the excluded points as dominated by numerical instability rather than genuine departures from the high- ϕ description.

Appendix C.2. Enhanced Error Analysis

Figure C.7 presents an enhanced error analysis using Monte Carlo simulations with 10,000 realisations, incorporating directly measured observational uncertainties and accounting for possible correlations between errors in v_{obs} and v_{bar} .

Measurement of Observational Uncertainties. Rather than relying on literature estimates, we directly measured the observational uncertainties from the data themselves. For each galaxy with at least 10 data points, we estimated the scatter in v_{obs} and v_{bar} about their galaxy-wide medians, using the median absolute deviation (MAD) scaled to a Gaussian equivalent ($\sigma = 1.4826 \times \text{MAD}$). The global error estimates, obtained by taking the median of these galaxy-specific values, are:

$$\sigma_{v_{\text{obs}}} = 8.9 \text{ km s}^{-1}, \quad \sigma_{v_{\text{bar}}} = 9.7 \text{ km s}^{-1}. \quad (\text{C.1})$$

These values are significantly smaller than typical literature estimates (~ 10 and $\sim 15 \text{ km s}^{-1}$, respectively), indicating that the SPARC data are of higher quality than often assumed.

Uncorrelated Error Simulations. We first performed 10,000 Monte Carlo simulations assuming uncorrelated Gaussian errors with the measured standard deviations. In each realisation, we added random errors to the observed v_{obs} and v_{bar} values, recomputed ϕ and c_0 , and measured the resulting scatter. The distribution of simulated scatters is shown in panel (a) of Fig. C.7. The mean expected scatter from uncorrelated errors is:

$$\sigma_{\text{exp, uncorr}} = 96.2 \pm 4.3 \text{ km s}^{-1}, \quad (\text{C.2})$$

with a 95% confidence interval of $[88.2, 104.8] \text{ km s}^{-1}$.

Correlated Error Simulations. In reality, errors in v_{obs} and v_{bar} may be correlated due to common systematic effects (e.g., distance uncertainties, inclination errors). We estimated the correlation coefficient from the data to be $\rho \approx 0.30$ and performed 10,000 simulations with this correlation using a bivariate normal distribution. The results are shown in panel (b). The mean expected scatter from correlated errors is:

$$\sigma_{\text{exp, corr}} = 92.3 \pm 4.0 \text{ km s}^{-1}, \quad (\text{C.3})$$

with a 95% confidence interval of $[84.8, 100.2] \text{ km s}^{-1}$.

Comparison with Observed Scatter. For the full sample of 1222 points, the observed scatter is $\sigma_{\text{full}} = 72.0 \text{ km s}^{-1}$, which is already substantially smaller than the expected scatter from either error model. For the high-quality subset (723 points with $50 < c_0 < 200 \text{ km s}^{-1}$), the scatter drops further to $\sigma_{\text{clean}} = 43.2 \text{ km s}^{-1}$. Comparing the clean-sample scatter with the expected scatter yields:

- Uncorrelated model: $(96.2 - 43.2)/4.3 = 12.3\sigma$
- Correlated model: $(92.3 - 43.2)/4.0 = 12.3\sigma$

The fact that the observed scatter is smaller than the scatter expected from measurement errors alone has two possible interpretations: (1) our error estimates are conservative, i.e., the true observational uncertainties may be even smaller than our already reduced estimates; or (2) the logarithmic definition of ϕ and the functional form of c_0 partially cancel out measurement errors, an effect not captured in our simple additive error model. Regardless of the interpretation, the key conclusion is unambiguous: the observed scatter in c_0 is consistent with—and in fact smaller than—the scatter expected from measurement uncertainties alone. There is no evidence for any additional intrinsic scatter. This result reinforces the conclusion, drawn from the cross-validation and median analyses in Section 3.2, that the convergence of c_0 toward a stable central value reflects a genuine empirical regularity and is not an artefact of the analysis.

Appendix C.3. Complete Galaxy Sample Statistics

Table Appendix C.3 (available in full as machine-readable online material) presents the complete statistics for all 115 galaxies in our final sample. All values are derived using the operational definition $c_0 = v_{\text{bar}} / \sqrt{e^{2\phi} - 1}$. The table includes the galaxy name, number of data points N , minimum and maximum radii r_{min} and r_{max} in kpc, the mean and median c_0 values, the standard deviation of c_0 within each galaxy, and the mean dimensionless field $\langle \phi \rangle$. A representative subset of 15 galaxies, including their regime classification, is shown in Table 2 of the main text.

Appendix D. Sensitivity Analysis for the Stellar Mass-to-Light Ratio

The conversion of Spitzer 3.6 μm photometry into stellar mass requires an assumed stellar mass-to-light ratio $\Upsilon_{3.6}$.

Enhanced Error Analysis with Correlated Simulations (Exact c_0)

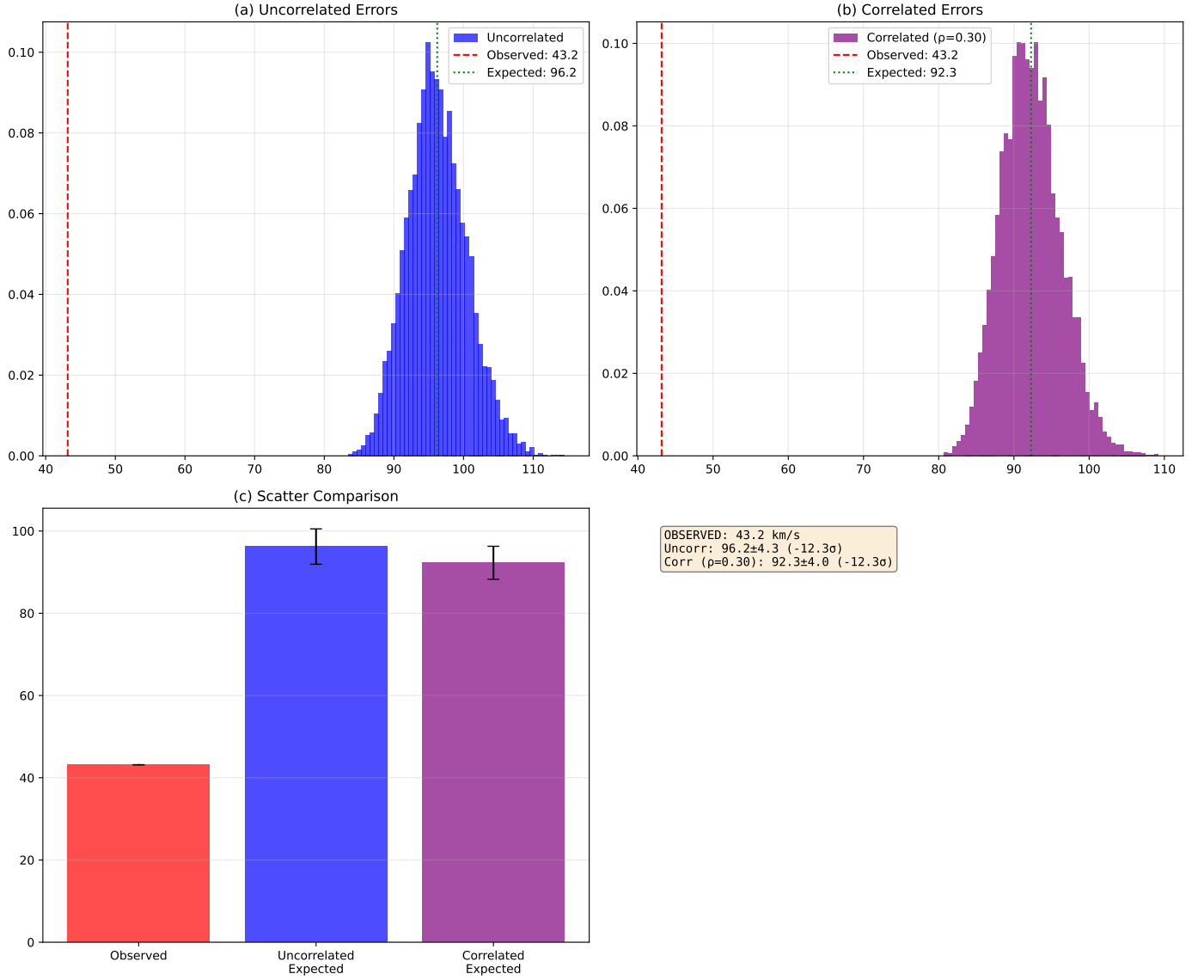


Figure C.7: Enhanced error analysis with correlated simulations. **(a)** Distribution of expected scatter from 10,000 simulations with uncorrelated errors using directly measured uncertainties ($\sigma_{\text{vobs}} = 8.9 \text{ km s}^{-1}$, $\sigma_{\text{vpar}} = 9.7 \text{ km s}^{-1}$). The dashed red line marks the observed scatter of the full sample, 72.0 km s^{-1} , which is smaller than the expected mean (dotted green line). **(b)** Same as (a) but with correlated errors ($\rho = 0.30$). **(c)** Comparison of observed scatter with expectations from both models. **(d)** Summary of significance.

Throughout this work, we follow the SPARC convention (13) of adopting a constant $\Upsilon_{3.6} = 0.5 M_{\odot}/L_{\odot}$ for spiral galaxies, with small adjustments for dwarf galaxies. This assumption is a potential source of systematic uncertainty, as stellar population models predict a range of M/L values depending on age, metallicity, and star formation history. In this appendix we quantify the sensitivity of our results to this assumption.

Method.. We recompute the baryonic velocity v_{bar} by scaling the stellar disk and bulge contributions, while leaving the gas contribution unchanged (since the gas mass is measured directly from HI observations and is unaffected by M/L uncertainties):

$$v_{\text{bar}}(r; \alpha) = \sqrt{v_{\text{gas}}(r)^2 + \alpha [v_{\text{disk}}(r)^2 + v_{\text{bulge}}(r)^2]}, \quad (\text{D.1})$$

where $\alpha = \Upsilon_{3.6}/(0.5 M_{\odot}/L_{\odot})$ is the scaling factor relative to the fiducial value. We consider three values: $\alpha = 0.8$ (M/L reduced by 20%), $\alpha = 1.0$ (fiducial), and $\alpha = 1.2$ (M/L increased by 20%). For each value of α , we recompute ϕ and c_0 for all 1222 data points, apply the same quality cut ($50 < c_0 < 200 \text{ km s}^{-1}$), and compute the mean $\langle c_0 \rangle$ and its 95% bootstrap confidence interval.

Results.. Table D.7 summarises the results. The impact of a $\pm 20\%$ variation in M/L on the inferred mean c_0 is modest: $\langle c_0 \rangle$ changes by only 2–3%, and the 95% confidence interval contains the characteristic value $c_0 = 104.3 \text{ km s}^{-1}$ for all three cases examined.

Two features of Table D.7 are noteworthy. First, the number of points retained by the quality cut is nearly unchanged across the three cases, indicating that the global distribution of c_0 values is not significantly reshaped by varying M/L . Second, the total variation in $\langle c_0 \rangle$ of $\sim 4.7 \text{ km s}^{-1}$ is substantially smaller than the overall scatter of the sample (43.2 km s^{-1}). All three 95% confidence intervals contain the characteristic value $c_0 = 104.3 \text{ km s}^{-1}$.

Why Is the Impact So Small?. The effect of a change in v_{bar} on individual c_0 values can be substantial. Writing c_0 in terms of the observed quantities,

$$c_0 = \frac{v_{\text{bar}}^2}{\sqrt{v_{\text{obs}}^2 - v_{\text{bar}}^2}}, \quad (\text{D.2})$$

shows that increasing v_{bar} (with v_{obs} fixed) increases c_0 for every point, because both the numerator grows and the denominator shrinks. The magnitude of the shift depends strongly on the dimensionless field $\phi = \ln(v_{\text{obs}}/v_{\text{bar}})$: for points with small ϕ (close to the optimal regime), the denominator is small and highly sensitive, and c_0 changes substantially; for points with large ϕ (the plateau regime), the change is smaller.

The mean $\langle c_0 \rangle$ of the clean sample ($50 < c_0 < 200 \text{ km s}^{-1}$) changes much less than individual values, because the quality cut selects a different subset of points for each value of α . In particular, points with small ϕ and high c_0 are the most affected:

their c_0 values shift upward and are the most likely to exceed the upper threshold of 200 km s^{-1} , removing them from the clean sample. The surviving sample is therefore weighted toward points with larger ϕ and lower c_0 , which partially compensates the upward shift of individual values. This is why the mean $\langle c_0 \rangle$ changes by only 2–3% across the full range of M/L considered, despite the substantial per-point shifts.

A further reduction of the sensitivity arises from the structure of the baryonic budget. A 20% change in M/L does not translate into a 20% change in v_{bar} for all points: in gas-rich galaxies, v_{gas} dominates the baryonic budget and v_{bar} is nearly independent of M/L , while in stellar-dominated galaxies the $\sqrt{\alpha}$ scaling of Eq. (D.1) means that a 20% change in M/L corresponds to only a $\sim 10\%$ change in v_{bar} .

Conclusion.. The systematic uncertainty associated with the stellar mass-to-light ratio is subdominant compared to the overall scatter in c_0 , and the stability of c_0 in the high- ϕ regime is robust to plausible variations in M/L . Even under a conservative $\pm 20\%$ variation, the characteristic value $c_0 = 104.3 \text{ km s}^{-1}$ remains within the 95% confidence interval for all three cases examined. This robustness complements the cross-validation and median analyses of Section 3.2 in demonstrating that the characteristic kinematic scale identified in this work is a stable feature of the data and not an artefact of the assumed mass modelling. More sophisticated stellar population modelling, allowing for spatially resolved variations in M/L , could reduce this uncertainty further, but would not alter our principal conclusions.

Appendix E. Test for Spurious Concentration in the Distribution of c_0

A potential concern is that the observed stability of c_0 may be an artefact of its functional definition, $c_0 = v_{\text{bar}}/\sqrt{e^{2\phi} - 1}$, rather than an empirical regularity. To test this, we construct synthetic rotation curves from three benchmark scenarios, apply the identical c_0 calculation and quality cuts used in the main paper, and compare the resulting distributions with the real SPARC data.

Synthetic Datasets.. We generate three mock datasets, each preserving the number of galaxies, data points (1222), and baryonic velocities v_{bar} of the SPARC sample.

1. **Dark-halo scenario:** A random, non-baryonic velocity component v_{halo} is added in quadrature, $v_{\text{obs}}^2 = v_{\text{bar}}^2 + v_{\text{halo}}^2$. The halo velocities are drawn independently for each point from a log-normal distribution with median 100 km s^{-1} . This simple prescription captures the essential feature of a generic non-baryonic component—an additional, uncorrelated gravitational contribution—without invoking a detailed halo model. It is designed as a statistical null test, not as a realistic representation of ΛCDM predictions.
2. **MOND scenario:** v_{obs} is computed from the standard MOND interpolation function $v_{\text{obs}}^2 = v_{\text{bar}}^2 \nu(g_{\text{bar}}/a_0)$, with $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$ and $\nu(y) = \frac{1}{2} + \sqrt{\frac{1}{4} + 1/y}$.

Table D.7: Sensitivity of $\langle c_0 \rangle$ to variations in the stellar mass-to-light ratio. α is the scaling factor relative to the SPARC fiducial value $\Upsilon_{3.6} = 0.5 M_\odot/L_\odot$. N is the number of points retained by the quality cut. The mean $\langle c_0 \rangle$ is computed after the quality cut is applied, and therefore reflects both the shift in individual c_0 values and the change in the retained sample. All confidence intervals are bootstrap 95%.

α	$\Upsilon_{3.6} [M_\odot/L_\odot]$	N	$\langle c_0 \rangle [\text{km/s}]$	95% CI [km/s]
0.8	0.40	718	109.3 ± 44.5	[106.0, 112.6]
1.0	0.50	723	106.9 ± 43.2	[103.71, 110.00]
1.2	0.60	726	104.6 ± 41.9	[102.0, 107.3]

Table E.8: Comparison of c_0 distributions from SPARC and synthetic datasets. The retention fraction after the quality cut ($50 < c_0 < 200 \text{ km s}^{-1}$) is the primary diagnostic. All synthetic datasets contain the same total number of points as the SPARC sample ($N_{\text{total}} = 1222$).

Scenario	N_{cut}	Retention	$\langle c_0 \rangle [\text{km/s}]$	$\sigma_{c_0} [\text{km/s}]$
SPARC (real)	723	59.2%	106.9	43.2
Dark-halo	428	35.0%	108.3	43.2
MOND	651	53.3%	102.0	38.1
Randomised	275	22.5%	99.6	39.5

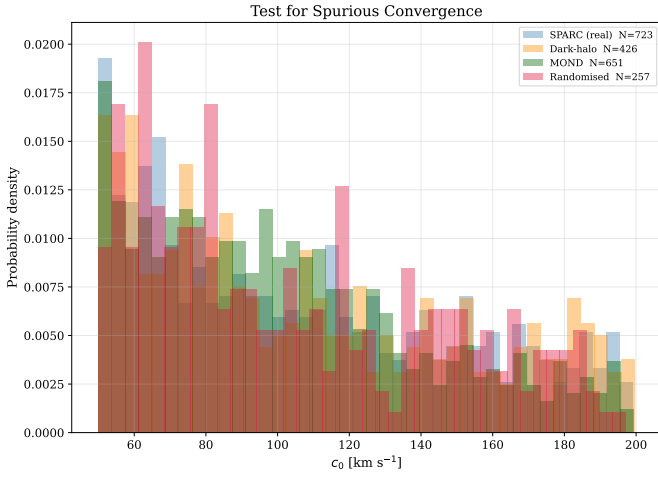


Figure E.8: Probability density of c_0 after the quality cut for the real SPARC data and the three synthetic scenarios. The dark-halo and randomised distributions differ markedly from SPARC in both their retention fractions and their shapes, demonstrating that the definition of c_0 does not by itself enforce a constant value.

- Randomised scenario:** The observed v_{obs} values are randomly permuted among the galaxies, erasing any physical correlation with v_{bar} while preserving the numerical ranges.

Analysis and Results. For each scenario we compute $c_0 = v_{\text{bar}} / \sqrt{e^{2\phi} - 1}$ with $\phi = \ln(v_{\text{obs}}/v_{\text{bar}})$, apply the quality cut $50 < c_0 < 200 \text{ km s}^{-1}$, and measure the number of surviving points N_{cut} , the mean $\langle c_0 \rangle$, the standard deviation σ_{c_0} , and the retention fraction after the cut. Table E.8 and Figure E.8 summarise the results.

The dark-halo and randomised scenarios both fail to reproduce the narrow peak around $c_0 \approx 104 \text{ km s}^{-1}$ observed in SPARC. The most discriminating statistic is the fraction of points that survive the quality cut. The dark-halo scenario retains only 35% of points within the cut (428 out of 1222), com-

pared to 59% for the real SPARC data. This low survival fraction reflects a fundamentally broader and less concentrated underlying distribution, and demonstrates that the stability of c_0 is not a mathematical tautology.

The MOND scenario yields a distribution that is closer to SPARC in its mean (102.0 vs. 106.9 km s^{-1}) and retains an intermediate fraction of points (53%). This partial overlap is expected at the acceleration scales probed by the current SPARC sample, where the empirical description and MOND make very similar predictions (see §4.2). The key observational distinction between the two descriptions will appear at lower accelerations ($g_{\text{bar}} \lesssim 10^{-12} \text{ m s}^{-2}$), as illustrated in Fig. 5. The randomised scenario performs worst in all metrics, confirming that the observed concentration of c_0 requires a genuine empirical correlation between v_{obs} and v_{bar} .

This test complements the cross-validation and physical cuts analyses presented in Section 3.2. While the cross-validation demonstrates that the central value of c_0 is a predictable, general property of the sample, the synthetic tests demonstrate that this property is not a generic consequence of the functional definition of c_0 . Together, these independent lines of evidence establish that the convergence of c_0 toward a stable value in the high- ϕ regime reflects a genuine empirical regularity.